

From Prompt to Portfolio: Generative AI and Retail Investment^{*†}

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Abstract

Generative AI has become one of the most rapidly adopted technologies, and this adoption extends to financial advice. Do individuals act on this advice? I study the staggered adoption of generative AI for financial advice by retail investors and its effects on their investment decisions and portfolio allocations. AI advice shifts new investment away from individual stocks and toward low-cost index funds, with corresponding changes in portfolio holdings. Using data linking investors' prompts and AI answers to their transactions and portfolio holdings, I find high rates of advice implementation.

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1 Introduction

Generative AI adoption is spreading at historic speed, outpacing that of the personal computer and the internet (Bick et al., 2026; Lynch, 2026). This adoption extends to financial advice. According to a recent Investing.com survey, nearly two-thirds of U.S. retail investors have used generative-AI tools to assist with investment decisions (Investing.com 2026).

Do individuals act on the advice they receive from generative AI? Current research on generative AI focuses on what people ask (Chatterji et al., 2025; Handa et al., 2025) and on evaluating AI-generated answers (Patwardhan et al., 2025; Appel et al., 2026; Ngangoue et al., 2026). In the context of financial advice, Choukhmane et al. (2026) collect questions that survey participants would ask AI, evaluate the recommendations AI provides in response, and simulate the effects of following that advice.

In this paper, I use data that links retail investors' prompts and generative AI answers to their transactions and portfolio holdings, provided by Mezzi, a wealth-management platform serving U.S. retail investors. To my knowledge, this is the first study to examine transaction- and portfolio-level evidence on the effects of generative AI financial advice. Using a local LLM to encode the advice supplied in response to investor-initiated prompts, I find that buy recommendations are concentrated in low-fee, diversified index funds. The linked data allow me to match AI recommendations to investors' subsequent transactions, and I find high rates of advice implementation.

Identifying the causal impact of generative AI advice is challenging because investors choose whether and when to seek advice, as well as what to ask about. I address this identification challenge using two complementary identification strategies.

First, I study the staggered adoption of generative AI for financial advice by retail investors and its effects on their investment decisions and portfolio allocations, following the introduction of a generative AI financial assistant on Mezzi in November 2024. I find that investors shift new investment away from individual stocks and toward these funds after adoption, with corresponding changes in portfolio holdings. The idiosyncratic risk of new purchases is lower following adoption. Second, I complement the staggered difference-in-differences design with a difference-in-differences design exploiting platform-side changes in AI advice.

I have three main findings.

First, I find high rates of implementation of generative AI advice. The advice predominantly recommends funds: 85.0 percent of ticker-level buy recommendations concern ETFs or mutual funds, compared with 15.0 percent for individual stocks. This pattern is consistent across time and models. Linking ticker-level recommendations to subsequent transactions, I find that 36.2 percent of buy-advice answers are followed by a purchase of at least one recommended security within ten trading days. Further, at the end of my sample period, 30.6 percent of investors hold at least one security introduced to them by generative AI.

Second, the primary staggered-adoption design finds that investors shift toward passive funds following adoption, with a 7.60-percentage-point increase in their share of purchases and a 2.73-percentage-point increase in their share of month-end equity holdings. I find no evidence that this shift is offset by riskier individual-stock purchases; the idiosyncratic risk of new purchases is lower following adoption. Purchased funds also have lower expense ratios. This result is robust to alternative staggered-adoption estimators,

passive-fund definitions, purchase-activity requirements, and adoption-cohort groupings.

Investors' allocation to funds changes with exogenous shifts in AI recommendations. Using a difference-in-differences design, I find that the fund share of purchases increases among exposed investors relative to non-users when ticker-specific buy recommendations become more frequent. This complements my main staggered-DID design.

Taken together, the two designs show that generative AI shifts retail investment, and that overall, it improves diversification by reducing individual-stock exposure and increasing the share in diversified funds.

Third, I study the mechanisms that lead to this shift. Bhattacharya et al. (2012) show that even free, unbiased financial advice has low take-up and limited implementation. I then investigate which mechanisms and features of generative AI advice may help overcome the low implementation documented for other forms of financial advice. Implementation is higher when answers are more personalized. These findings suggest that personalized advice may help investors recognize concentration risk in their existing holdings and direct new investment toward diversified funds. I also find suggestive evidence of learning: investors are more likely to use financial concepts in subsequent prompts after generative AI introduces them.

Related literature. First, I contribute to the emerging literature on the economic and financial effects of generative AI. Mo and Ouyang (2025) provide a comprehensive review of this literature. Brynjolfsson et al. (2025) find productivity gains among customer-support workers, especially those with less experience. Related work examines writing tasks (Noy and Zhang, 2023) and investors' information processing (Blankespoor et al., 2026). OpenAI and Anthropic studies document what people ask and the tasks for which they use generative AI (Chatterji et al., 2025; Handa et al., 2025). Choukhmane et al. (2026) ask a representative sample to write prompts seeking financial advice from LLMs and simulate the effects of following the advice. They find that following this advice would lead to choices more consistent with life-cycle theory, including greater participation in diversified equity funds. Ngangoue et al. (2026) find that ChatGPT's advice on major life decisions varies with socioeconomic background. Other work uses LLMs as simulated economic agents (Horton et al., 2023; Hao and Xie, 2025; Verstyuk and Douglas, 2025), examines their preferences and biases (Ross et al., 2024; Bini et al., 2026; Ouyang et al., 2025; Fedyk et al., 2026), and studies how financial advice responds to investor characteristics (Rumpf et al., 2026; Fieberg et al., 2025; Takayanagi et al., 2025; Foltyn and Olsson, 2026). My contribution is to provide, to my knowledge, the first evidence on the implementation and effects of AI advice.

Second, I contribute to the literature on financial advice by examining a new form of financial advice. Existing literature focuses on human financial advisors and robo-advice. The motivation behind this literature is that, in theory, financial advice might mitigate under-diversification and other mistakes observed in retail investors. However, in practice, the literature has identified several shortcomings of human advice. First, incentives may distort advice (Mullainathan et al., 2012; Chalmers and Reuter, 2020). Further, Bhattacharya et al. (2012) show that even unbiased and beneficial advice is rarely followed. They offer advice to a sample of German households and show that few households follow the provided advice. Other work examines advisor beliefs and limited tailoring (Linnainmaa et al., 2021; Reuter and Schoar, 2024), including Canadian evidence that clients' portfolios closely reflect their advisers' own asset allocations (Foerster et al., 2017). The literature on robo-advice finds more promising evidence of its benefits (D'Acunto et al., 2019; Rossi

and Utkus, 2024; see D’Acunto and Rossi, 2023, for a review). Generative AI advice differs from automated robo-advice in implementation: individuals must choose which recommendations to follow and execute the trades themselves. Conversational evidence offers an alternative to survey-based methods for studying investors’ beliefs and portfolio choices (e.g., von Gaudecker, 2015; Choi and Robertson, 2020; Giglio et al., 2021). I find that implementation is higher when answers are more personalized, including when they use information from linked accounts. This finding complements evidence that demographic similarity increases adherence to human advice (Stolper and Walter, 2019). I also find suggestive evidence of learning: investors are more likely to use financial concepts in later prompts after generative AI introduces them.

Finally, I contribute to the literature on household investment mistakes more broadly, and on what forms of treatment can improve financial decisions (see Campbell, 2006 for a review). This literature documents underdiversification (Calvet et al., 2007; von Gaudecker, 2015), excessive trading (Barber and Odean, 2000), and costly fund selection (Choi et al., 2010; Bailey et al., 2011; Bhattacharya et al., 2017). Explanations for underdiversification, and in particular the combination of single stocks with some weight in diversified funds, include probability weighting (Polkovnichenko, 2005; Dimmock et al., 2021) and incentives to specialize in learning about particular assets (Van Nieuwerburgh and Veldkamp, 2010). Other studies examine financial inertia (Madrian and Shea, 2001; Andersen et al., 2020) and the effects of financial education (Fernandes et al., 2014). I show that AI advice has the potential to mitigate underdiversification and reduce fund expenses.

The remainder of the paper proceeds as follows. Section 2 describes the platform and linked data. Section 3 documents what investors ask and what the AI recommends. Section 4 presents the empirical strategy. Section 5 reports the main results and robustness checks. Section 6 studies implementation and mechanisms. Section 7 concludes.

2 Setting and Data

This section describes the data I analyze. I first describe the Mezzi platform and linked financial-account records, and then explain how I construct portfolio and conversation measures.

2.1 Firm Background

I use data on U.S. retail investors from Mezzi, a wealth-management platform that launched as a portfolio-tracking application in November 2023. In November 2024, Mezzi introduced a generative AI-based conversational assistant that provides financial advice to its customers. Investors interact with generative AI through a chat interface, submitting prompts about their investments and receiving LLM-generated responses. Generative AI does not execute trades; investors decide whether and how to implement the advice they receive.

Mezzi integrates generative AI with investors’ financial-account data through third-party aggregators, covering checking accounts, brokerage accounts, and tax-advantaged accounts such as 401(k)s, Roth/IRA accounts, and 529 accounts. To ensure a complete financial picture of its investors, Mezzi provides onboarding calls and assists investors with connecting brokerage, retirement, and other accounts.

Existing investors and newer sign-ups to the app subsequently adopted generative AI. I define adoption as the date of an investor’s first observed prompt.

2.2 Financial and Conversation Data

Through third-party aggregators, Mezzi obtains the transaction histories of investors' linked accounts.¹ Transactions and dividend receipts are recorded on a daily basis. For each transaction, I observe the date, security identifier, transaction type, quantity, price, dollar amount, and fee. The records include purchases, sales, transfers, and reinvestments and extend back up to two years before investors first connect their accounts.

Information such as prices and returns comes from CRSP and CRSP Mutual Funds.² I match the securities in the account records to the CRSP stock and mutual fund databases. The account data, in conjunction with the CRSP market data, allow me to compute daily portfolio holdings, values, and returns.

I first infer daily holdings from transactions, transfers, reinvestments, and dividend receipts. When a position predates the available transaction history, I recover the number of units held by dividing the cash dividend or fund distribution recorded in the account by the corresponding CRSP distribution per share. Purchases, sales, reinvestments, and split adjustments are then used to update these holdings. I value each position at the daily closing price and aggregate holdings across linked accounts.

In addition to transaction and dividend data, Mezzi periodically obtains snapshots of holdings data, which are used to validate the thus-constructed portfolios. The constructed daily portfolio snapshots contain market values, asset-class composition, concentration measures, and returns.

The observed portfolio combines the accounts linked to Mezzi. Assets in unlinked accounts are outside these records, so the portfolio weights describe the investor's connected accounts.

I measure purchase shares relative to total purchase dollars, and the passive-fund holdings share relative to month-end equity holdings. Appendix B provides the detailed variable definitions.

The classified conversational sample contains 69,598 initial and follow-up prompts and their corresponding AI answers, with a median of 12 prompts per investor.

2.3 Summary Statistics

Table A.1 provides the key summary statistics for the main sample. Panel A reports sample size and demographic characteristics, and Panel B reports portfolio characteristics. For comparison, the table also reports the corresponding statistics for SCF respondents with equity holdings in the 2022 Survey of Consumer Finances. Table A.2 compares monthly turnover of Mezzi investors with that in Giglio et al. (2021) and Barber and Odean (2000). An important distinction of the Mezzi data is that they track investments across accounts at different institutions, while Giglio et al. (2021) and Barber and Odean (2000) use accounts at Vanguard and one discount brokerage, respectively. When I restrict the Mezzi data to each investor's largest account to make the estimates more comparable, Mezzi users have higher mean monthly turnover than Vanguard investors (7.50 versus 2.30 percent), but turnover comparable in magnitude to the purchase turnover reported by Barber and Odean (7.50 versus 6.49 percent).

Most Mezzi equity owners combine indirectly held equity with substantial investments in individual

¹Before being made available for this research, the records are stripped of personally identifiable data. All analyses are conducted on these de-identified records, and only aggregate results are reported.

²Security identifiers are supplemented using the CRSP–Compustat linking table; historical option prices are from HistoricalOptionData.com.

stocks. Among direct-stock owners, the median investor holds 13 stocks and allocates 41.0 percent of total equity to direct stocks. The median total-equity, indirect-equity, and direct-equity shares of the observed portfolio are 75.3 percent, 43.4 percent, and 25.3 percent, respectively. Applying the same definitions to the 2022 Survey of Consumer Finances yields corresponding medians of 35.9 percent for the direct fraction of equity, four directly held stocks, and portfolio shares of 44.0 percent, 37.8 percent, and 10.9 percent.

Mezzi investors lean wealthier than equity-owning SCF households (median observed portfolio value of approximately \$556,000 versus \$155,000) and are more active investors (75.0 percent versus 28.2 percent hold more than one individual stock). However, these characteristics are consistent with survey evidence on who uses GenAI for financial information. An HSBC–Ipsos survey finds that 82 percent of high-net-worth investors with at least \$2 million in investable assets use AI for finance and investment tasks, compared with 73 percent across its full sample of affluent and high-net-worth investors. Only 9 percent of high-net-worth respondents report that AI has not meaningfully changed how they manage their wealth, compared with 26 percent of investors with \$100,000–\$250,000 in investable assets (HSBC Holdings plc, 2026). Blankespoor et al. (2026) similarly find that more sophisticated retail investors lead adoption. Thus, while the Mezzi sample is not representative of the overall SCF population, its characteristics are consistent with those of the retail investors most likely to use GenAI for investment decisions.

3 The Content of Generative AI Financial Advice

This section examines the content of investors’ prompts and the investment advice they receive from generative AI. Figure A.1 shows word clouds for investors’ prompts and generative AI responses as a high-level overview of the topics discussed. Choukhmane et al. (2026) provide a comprehensive comparison of the topics and financial products discussed in survey respondents’ prompts and the corresponding LLM advice. I focus on ticker-level advice to inform the staggered-adoption design based on how advice differs from investor intentions revealed in prompts. First, I compare the securities mentioned in investors’ prompts with those recommended in gen AI answers. Second, I examine how generative AI buy and sell recommendations differ.

I identify stock and fund terms and explicit ticker symbols in the text. I use an LLM to identify buy and sell recommendations in gen AI answers. In addition, I categorize prompts as either open-ended or expressing a specific investment idea. This allows me to examine what generative AI recommends when investors have not proposed a specific investment idea.

The most pronounced difference between AI-recommended tickers and those investors express interest in is the greater emphasis on ETFs and funds in AI recommendations. Table A.3 compares buy interest in investors’ prompts with buy recommendations in gen AI answers. A message using both words and tickers of the same type counts once. Each column averages messages within investors and then gives each investor equal weight. Context-only ticker mentions in answers are excluded. Answers without recommendations remain in the denominator.

For the average investor, stocks are considered for purchase in 1.9 percent of prompts and recommended in 4.6 percent of gen AI answers. ETFs and funds are considered for purchase in 3.5 percent of prompts and recommended in 18.0 percent of answers. The prevalence of ETF and fund buy recommendations implies

that gen AI adoption could shift subsequent investment toward funds.

Another feature of gen AI advice is that buy recommendations predominantly concern ETFs and funds, while sell recommendations are more evenly divided between funds and individual stocks. Panel A of Table 1 reports the composition of ticker-level recommendations. For this table, open prompts are classified by the LLM as having no implied desired investment action—that is, they do not ask about a specific investment idea. Among buy recommendations responding to open-ended prompts, 89.7 percent concern an ETF or mutual fund and 10.3 percent concern an individual stock. Across all prompts, the corresponding shares are 85.0 and 15.0 percent. Panel A gives each ticker-level buy or sell recommendation equal weight. Appendix Table A.3 includes answers without buy recommendations and gives each investor equal weight. The fund tilt is therefore present even when the investor has not proposed a specific investment idea. Sell recommendations are nearly balanced between funds and individual stocks, with funds accounting for 49.6 percent.

Panel B measures how widely each type of advice is received. Within each category, a recipient is counted once regardless of how often that advice is repeated. Overall, 96.9 percent of investors in my sample who receive buy advice receive at least one ETF or fund buy recommendation, compared with 48.0 percent for individual stocks (Panel B of Table 1).

Table 1. Composition of ticker-level recommendations

	ETF/fund (%)	Individual stock (%)	Difference (pp)
<i>Panel A. Share of recommendations</i>			
Open prompts: buy	89.7	10.3	79.3***
Open prompts: sell	52.6	47.4	5.2
All prompts: buy	85.0	15.0	70.0***
All prompts: sell	49.6	50.4	-0.9
<i>Panel B. Share of investors receiving at least one recommendation of each type</i>			
Buy recipients	96.9	48.0	48.9***
Sell recipients	70.0	80.7	-10.7***

Notes: Panel A reports the security-type composition of exact ticker-level recommendations. Open prompts are classified by the LLM as having no implied desired investment action. Panel B reports the share of individuals receiving at least one recommendation of each type. An individual may receive both types, so recipient shares need not sum to 100. Difference is defined as the ETF/fund share minus the individual-stock share, calculated before rounding. The tests compare the two shares, using investor-clustered standard errors in Panel A and paired investor-level *t*-tests in Panel B. All tests are two-sided. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In a more general classification that also captures advice without a ticker, 10.1 percent of classified answers recommend adding to broad/index funds or diversified baskets. Conditional on a ticker being mentioned in the answer, this share is 14.2 percent.

4 Empirical Strategy

I identify the effect of AI financial advice using two complementary difference-in-differences designs. The first exploits staggered variation in when users begin interacting with the AI assistant. Users begin using the assistant between November 2024 and July 2026, and I observe up to two years of historical account data, enabling me to compare their investment behavior before and after adoption.

The second exploits two platform-side changes in the system prompt that made the advice more explicit and actionable, in the sense that the share of advice recommending tickers first increases and then decreases

dramatically.

The two designs are complementary in that they have different strengths and face different threats to identification. The staggered-adoption design draws on adoption events distributed across users and calendar dates, but users endogenously choose whether and when to adopt. The platform-change design holds users’ prior selection into AI use fixed, but is more exposed to contemporaneous developments because the system prompt changed on common calendar dates.

The results are consistent across the two estimation strategies. Both show that AI financial advice shifts investment away from individual stocks and toward diversified, passive, low-cost funds. The magnitudes also line up quantitatively across the two designs. Scaling the estimates by the corresponding increase in exposure to diversified-fund recommendations, both designs imply a similar increase in fund investment per additional fund/ETF recommendation.

4.1 Staggered Adoption

First, I use a staggered-adoption design exploiting the fact that the data provide historical transactions and portfolio details up to two years prior to a user signing up for Mezzi and starting to use the AI financial assistant.

Following the literature, I use the interaction-weighted estimator of Sun and Abraham (2021) to avoid the “forbidden comparison” problem that two-way fixed-effects estimators suffer from in staggered-adoption designs.

For individual outcomes, I exploit variation in the month of first use. The analysis retains individuals with a common October 2024 portfolio baseline; individuals with no observed AI prompt form the comparison group. For individual i in calendar month t , I estimate cohort-specific event-time effects around first use:

$$y_{it} = \alpha_i + \delta_t + \sum_g \sum_{k \neq -1} \beta_{gk} \mathbf{1}\{\text{AI}_i = g\} \mathbf{1}\{t - g = k\} + \varepsilon_{it}, \quad (1)$$

where AI_i is the first-adoption month and month -1 is omitted. I aggregate the cohort-specific coefficients using the interaction-weighted estimator of Sun and Abraham (2021) and report their average over event months 1 through 9. Observations are individual-months, and standard errors are clustered by individual and calendar month.

The main concern with this design is selection into treatment. AI usage is clearly not randomly assigned, and the timing of AI adoption as well as the questions asked of the AI financial assistant are endogenously chosen by users. Given this concern, I focus on an outcome variable that is consistently targeted by AI advice and is stable across time and variation in user questions and characteristics. The strongest time- and query-invariant feature of AI-generated advice is its tendency to recommend diversified funds rather than individual stocks. In my data, 89.7 percent of AI buy recommendations are for an ETF or mutual fund, compared with 10.3 percent for an individual stock. At the user level, 96.9 percent of users receiving buy advice receive at least one ETF or fund recommendation. This preference is also externally validated by Choukhmane et al. (2026), who use investment questions written by a representative survey sample, feed these questions to LLMs, and find that the advice would move respondents toward higher participation in

diversified equity funds.

Individual fixed effects absorb persistent differences across investors, and calendar-month fixed effects absorb shocks shared by the sample. The specification therefore compares within-individual changes around first use with common calendar-time movements.

My primary outcome is the share of purchase dollars allocated to passive index funds. I also examine the passive-index share of month-end equity holdings, net investment between funds and individual stocks, the idiosyncratic risk of purchased securities, and the fees of purchased funds.

4.2 Platform-Side Change in Recommendation Actionability

Second, I complement this design with a second DID study using a platform-side change in how advice is supplied.

Figure C.1 plots the shift in actionability, the raw ETF/fund purchase measures, and the differences between the two exposure groups. The share of advice containing explicit ticker buy recommendations increases from a weekly average of 18.0 percent before the change to 33.4 percent during the high-actionability period beginning in the calendar week of December 15, 2025, and then declines to 5.6 percent during the low-actionability period beginning in the calendar week of January 12, 2026.

This shift in treatment happens in a relatively short window of less than four weeks. To ensure a reasonable probability of receiving advice in the different treatment windows, I sort users on pre-September 2025 AI usage. The idea is that, given some degree of persistence in how much investors seek AI advice, the group with higher exposure has the best chance of receiving advice under the different regimes.

Thus, this avoids the endogenous treatment-timing selection problem of the staggered DID (Section 4.1). The identification assumptions are that only AI users are affected by the platform-side shift in advice and that, absent this shift, high-exposure users and permanent non-users would have followed parallel trends in investment outcomes. For gross purchase share, the staggered-DID estimates show a shift in purchases rather than an upward trend, thus supporting parallel trends after the initial response to adoption (Figure 1, Panel A). The ongoing treatment effect of using AI for financial advice must not generate differential trends over this period (de Chaisemartin and D’Haultfœuille, 2026).

My main regression specification for this design is a simple DID. The weekly outcome is the ETF and fund share of gross purchase dollars. The regression includes individual and calendar-week fixed effects:

$$y_{iw} = \alpha_i + \delta_w + \beta_H(AI_i \times \text{High}_w) + \beta_L(AI_i \times \text{Low}_w) + u_{iw}. \quad (2)$$

The coefficient β_H measures the change in the purchase-share gap between high-exposure investors and non-users from the pre-period to the high-actionability period. The change in that gap when actionability subsequently falls is $\beta_L - \beta_H$.

The pre-period is omitted. I report the highest-quartile differential during the high-actionability period and its change when actionability falls. Standard errors are clustered by individual and calendar week. Table C.1 reports the estimates, and Figure 2 reports the corresponding event studies.

5 Main Results

5.1 Main Outcomes

Figure 1 shows the Sun–Abraham event-study estimates for the impact of AI financial advice on the monthly purchase share and the month-end portfolio share. I find an immediate 7.08-percentage-point jump in the purchase share in the adoption month. The portfolio change is more gradual as purchases accumulate. The shift in the net passive-fund share is less pronounced, consistent with sale decisions also being driven by tax and rebalancing considerations.

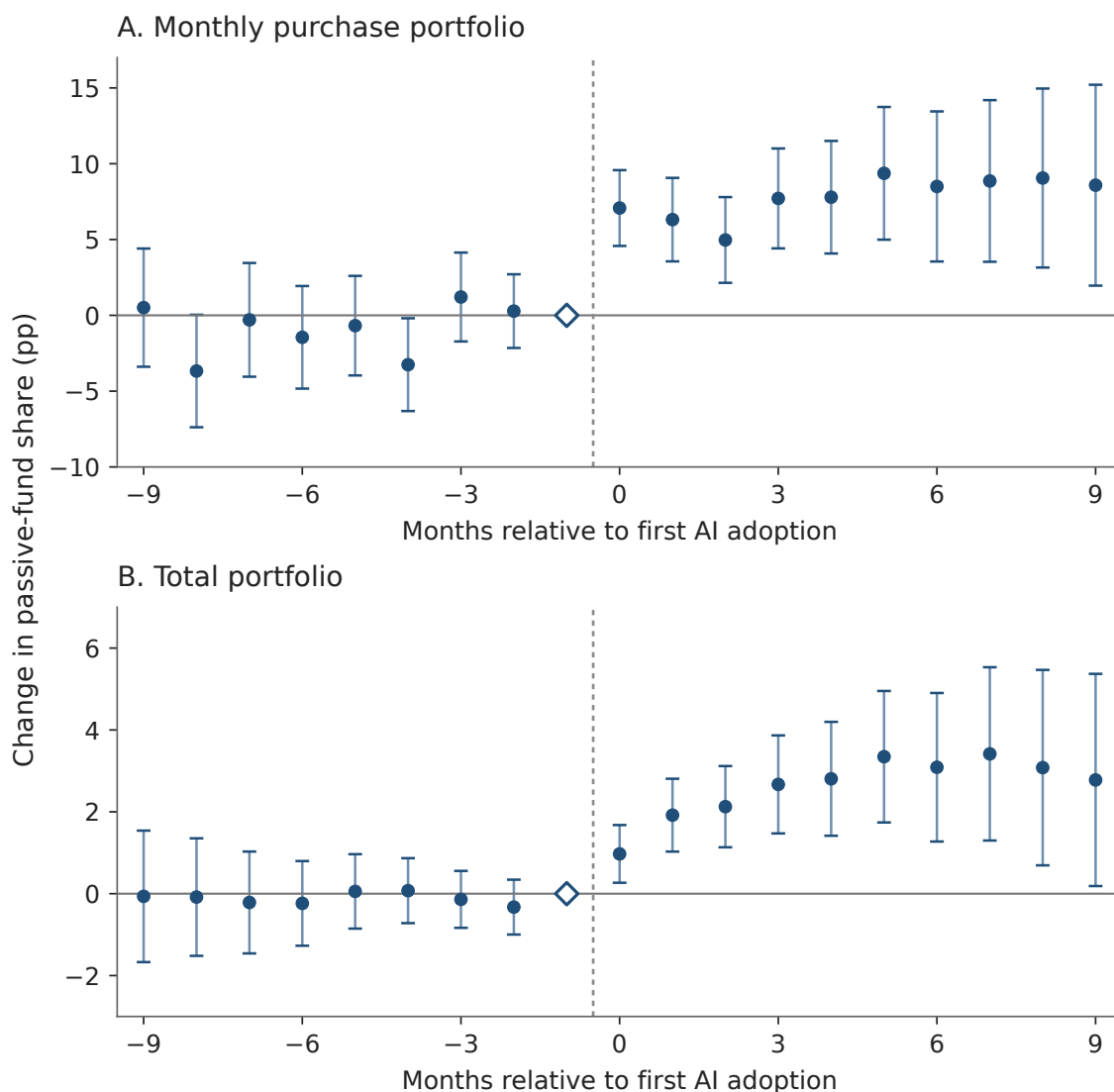


Figure 1. Passive-fund shares after AI adoption

Notes: This figure plots the estimated coefficients and 95 percent confidence intervals from Sun–Abraham event-study regressions. The dependent variable is the direct-index fund share of the monthly purchase portfolio in Panel A and the direct-index share of month-end equity holdings in Panel B. Event time is measured relative to AI adoption, and month -1 is omitted. Standard errors are double-clustered by individual and calendar month.

Table 2 shows the impact of AI adoption on investment behavior. Column (1) reports that, controlling for calendar-month and individual fixed effects, obtaining AI financial advice increases the share of monthly investments going to direct-index funds by 7.60 percentage points, or 22.8 percent relative to the dependent-variable mean of 33.30 percent.

Column (2) reports the estimated impact on the net passive-fund share, defined as fund purchases plus individual-stock sales minus fund sales and individual-stock purchases, scaled by gross noncash trading. The net passive-fund share increases by 4.03 percentage points. Column (3) reports the effect on direct-stock purchases: the individual-stock share falls by 5.88 percentage points. Finally, columns (4) and (5) report the impacts on month-end equity holdings: the direct-index fund share increases by 2.73 percentage points, and the individual-stock share falls by 2.53 percentage points.

Table 2. Effects of AI adoption on investment allocation

	Monthly purchase portfolio			Month-end portfolio holdings	
	Passive-fund share	Net passive-fund share	Individual-stock share	Passive-fund share	Individual-stock share
	(1)	(2)	(3)	(4)	(5)
Post-adoption effect	7.60*** (1.55)	4.03** (1.95)	-5.88*** (1.71)	2.73*** (0.68)	-2.53*** (0.71)
DV mean	33.30	14.39	46.81	38.39	49.69
R^2	0.61	0.36	0.63	0.90	0.91
Individual fixed effects	Yes	Yes	Yes	Yes	Yes
Calendar-month fixed effects	Yes	Yes	Yes	Yes	Yes

Notes: This table reports estimates from panel regressions of investment allocation on AI adoption. The unit of observation is an individual-month. The dependent variables in columns (1)–(5) are the direct-index fund share of purchases, the selling-inclusive net passive-fund share, the individual-stock share of purchases, the direct-index share of month-end equity holdings, and the individual-stock share of month-end equity holdings, respectively. Column (1) restricts the sample to months with purchases of at least 0.5 percent of the opening portfolio. No-trade months are missing in column (2). All columns report Sun–Abraham estimates. Standard errors are double-clustered by individual and calendar month and reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The increase in fund allocations leads to a complementary decrease in idiosyncratic volatility. This matters because AI adoption could increase the diversified part of the purchase portfolio while also increasing purchases of volatile individual stocks.

Table D.1 reports additional outcomes related to the characteristics of purchases. Columns (1)–(3) report different measures of idiosyncratic volatility. Shifting from individual stocks to diversified funds lowers idiosyncratic volatility: it falls by 13.6 percent using a U.S. market benchmark, 11.0 percent using the MSCI World Index, and 11.5 percent using Welch’s BSWA estimator. Column (4) reports the average volatility of stock-only purchases and shows no statistically significant change. Column (5) reports the expense ratio of funds purchased, which decreases by 3.57 basis points. Following Calvet, Campbell, and Sodini (2007), I also report the return loss of the combined monthly purchase portfolio. It falls by 39.9 percent.

Figure D.1 shows the distributions of purchase-portfolio idiosyncratic volatility, total volatility, the passive-fund share of purchases, and fund expense ratios before and after AI adoption. Figure D.2 reports the stacked-DID event-study estimates for the idiosyncratic volatility of purchases.

Conditional on purchasing individual stocks, I find no statistically significant change in their concentration, total volatility, or idiosyncratic volatility (Table D.2). The estimated effects on fund expense ratios remain negative across alternative estimators and when all fee-covered fund purchases are included, although precision varies across specifications (Table D.3).

These results show a decline in the idiosyncratic volatility of new purchases following AI adoption, consistent with the increase in fund investment. I find no evidence of an offsetting increase in the risk of individual-stock purchases.

Figure 2 and Table C.1 report the results for changes in recommendation actionability. An increase in actionability is associated with an 8.13-percentage-point increase in the ETF/fund share of gross purchases for high-exposure investors relative to permanent non-users. In contrast, this share falls by 7.67 percentage points when actionability decreases. The outcome includes all ETFs and funds, a broader category than the direct-index outcome in Table 2.

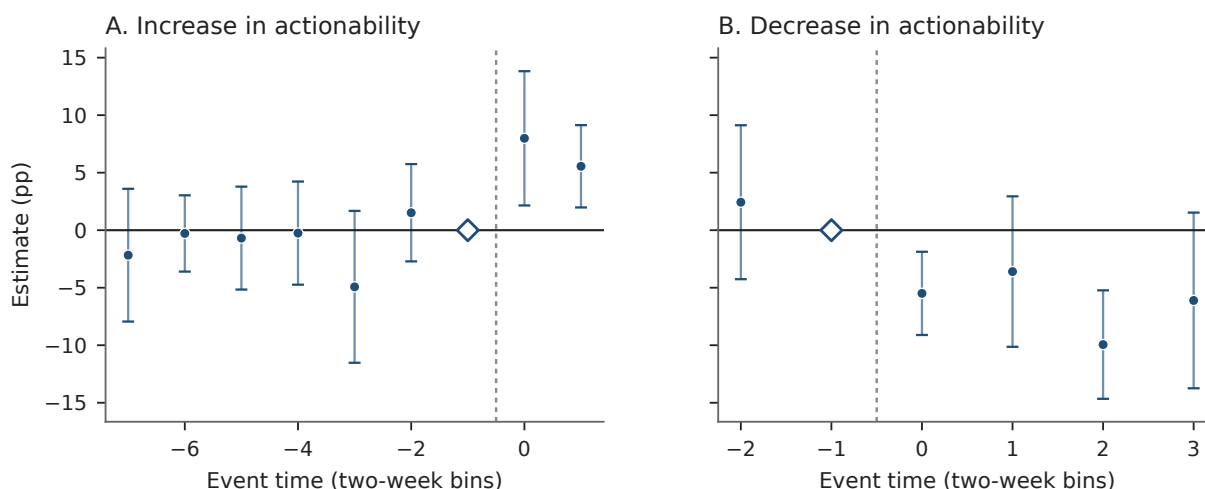


Figure 2. ETF/fund purchase shares around changes in recommendation actionability

Notes: This figure plots the estimated coefficients and 95 percent confidence intervals from event-study regressions around the increase and decrease in recommendation actionability. The dependent variable is the ETF/fund share of gross noncash purchase dollars. Event time is measured in bins of two complete calendar weeks, with bin -1 omitted. All regressions include individual and calendar-week fixed effects. Standard errors are double-clustered by individual and calendar week.

5.2 Heterogeneity by Portfolio Characteristics

The robo-advice literature shows that investors with less diversified portfolios tend to benefit more from advice (D'Acunto, Prabhala, and Rossi, 2019; Rossi and Utkus, 2024). I explore two dimensions in which this may apply to generative AI advice. First, I sort individuals into quintiles of pre-AI fund shares. Table 3 shows how the effect varies across quintiles. I find that the estimated effect of generative AI advice adoption on the fund share of purchases is strongest for the bottom two quintiles. This result is consistent with the

idea that individuals with lower initial fund shares have more scope to benefit from advice that encourages diversification.

Table D.4 reports the corresponding results using Sun–Abraham and stacked-DID specifications. The stacked-DID test does not reject the null hypothesis that the effects are equal across quintiles.

Table 3. Heterogeneity by pre-AI diversification

	Q1	Q2	Q3	Q4	Q5
<i>Passive-index-fund purchase share</i>					
Post-adoption effect	7.45***	14.35***	6.82***	3.99	1.04
	(1.81)	(2.53)	(2.49)	(2.68)	(1.98)

Notes: This table reports Sun–Abraham estimates separately by pre-AI diversified-fund portfolio-share quintile. Individuals are assigned to groups using this share, with assignments fixed once. The dependent variable is the direct-index fund share of the monthly purchase portfolio, measured in percentage points. Standard errors are double-clustered by individual and calendar month and reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

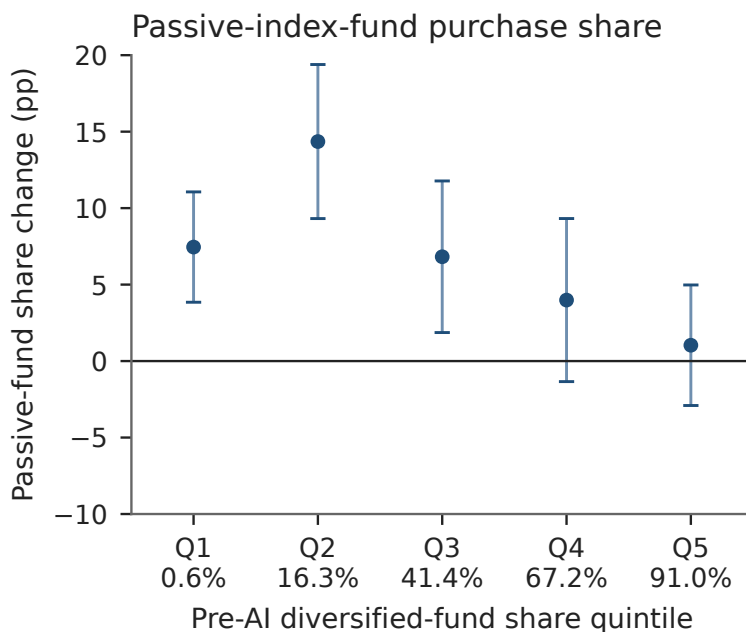


Figure 3. Effects by pre-AI diversified-fund share

Notes: This figure reports Sun–Abraham estimates separately by pre-AI diversified-fund portfolio-share quintile. Individuals are grouped according to this share, with quintile assignments fixed once. The dependent variable is the direct-index fund share of the monthly purchase portfolio, measured in percentage points. Each point is an estimated coefficient; the error bars indicate 95 percent confidence intervals. Standard errors are double-clustered by individual and calendar month.

5.3 Robustness

In this subsection, I conduct a number of robustness tests of the main results.

First, I show that my main result is robust to using different staggered difference-in-differences estimators. Table D.5 and Figure D.3 show the main estimates using Sun–Abraham, stacked DID, Callaway–Sant’Anna, Borusyak–Jaravel–Spiess, de Chaisemartin–D’Haultfoeulle, and traditional two-way fixed effects. The coefficients for the monthly purchase portfolio remain economically and statistically significant and of similar magnitude. The net passive-fund share estimates are also positive and statistically significant in three of the four available specifications. The estimates for the month-end portfolio share remain positive but are smaller and less precisely estimated.

Second, I examine a simple event study to observe the raw mean changes in the key variable following AI adoption. As with the staggered design, Figure D.4 shows no systematic pre-adoption trend and a clear increase in the fund share following adoption. Because the figure includes adopters only and does not adjust for calendar time or a comparison group, it is descriptive rather than a raw representation of the staggered difference-in-differences estimator.

Third, I examine robustness to different passive-fund definitions. Table D.6 reports estimates using the direct-index and CRSP B/D/E index-fund definitions. The estimates for both the monthly purchase portfolio and the month-end portfolio remain positive and similar under the narrow and broad definitions.

Fourth, I examine an alternative to the staggered DID estimation. Carlin et al. (2023) use the release of a mobile financial application in intention-to-treat and DID specifications to study how easier access to financial information affects consumer financial decisions. I replicate this approach by using the November 2024 introduction of the AI assistant on Mezzi as the common start-of-treatment date. I compare changes in fund purchase shares between investors who begin using AI within nine months of launch and investors who do not. The plot in Appendix Figure D.5 indicates parallel trends before users start adopting AI, followed by an increase in fund purchase shares among AI users relative to non-users. Panel A plots cumulative AI adoption. Panel B plots the weekly fund purchase shares for the two groups, and Panel C plots the difference between them. Panel D reports the DID estimates. Following the rollout, fund purchase shares increase by 7.89 percentage points among AI users relative to non-users.

Fifth, I examine robustness to the minimum activity required in a month. Table D.7 reports estimates using every positive purchase month and purchase requirements of 0.1 and 0.5 percent of the opening portfolio. The estimated increase in the passive-fund purchase share remains positive and statistically significant under each requirement.

Finally, I show that the average treatment effect is not skewed by a single adoption cohort. Table D.8 and Figure D.6 report treatment effects for early, middle, and late adopters under two cohort definitions. The first defines cohorts by dividing the months since November 2024 when the AI assistant was introduced into equal periods. This keeps the duration of the adoption windows fixed but allows the number of adopters to vary. The second defines cohorts by dividing into groups of approximately equal numbers of adopters. This balances cohort size but allows the calendar span to vary. Although later cohorts exhibit somewhat larger point estimates, the estimates are positive across both groupings, and the equality tests do not reject the hypothesis that the three cohort effects are the same.

6 Implementation and Mechanisms

This section investigates the mechanisms that lead to the shift toward funds following generative AI adoption. Bhattacharya et al. (2012) show that even unbiased, high-quality advice is not necessarily implemented. I show direct evidence of advice implementation, with shifts toward funds consistent with the staggered-adoption findings. I therefore investigate which mechanisms and features of generative AI advice may help overcome the low implementation documented for other forms of financial advice.

Section 6.1 documents high rates of advice implementation. Section 6.2 examines how advice personalization relates to implementation and finds that implementation is higher when answers are more personalized. These findings suggest that personalized advice may help investors recognize concentration risk in their existing holdings and direct new investment toward diversified funds. Finally, Section 6.3 provides suggestive evidence of learning: investors are more likely to use financial concepts in subsequent prompts after generative AI introduces them.

6.1 Implementation and Portfolio Accumulation

My main results in Section 5 show that, on average, retail investment shifts away from single stocks towards passive index funds following the adoption of generative AI advice. I supplement these monthly estimates with high-frequency, ticker-level evidence on the implementation of particular recommendations. Using a local LLM, I identify ticker-specific buy and sell recommendations in each response; Appendix E reports the classification prompt. These ticker-level recommendations are then linked to investments made in the following 10 days.

Table F.3 reports the rate at which investors implement AI recommendations within 1, 2, 5, and 10 trading days. 36.2 percent of answers with buy-advice are implemented within 10 days. At the ticker-level, 18.5 percent of recommended securities are purchased. 72.6 percent of answers containing a ticker-level buy recommendation recommend more than one ticker, and thus the implementation rate at the answer level is higher.

To compare the recommended securities with other contemporaneous investments, I construct two portfolios for each advice event. The first consists only of the recommended securities. The second consists of the “non-compliant” investments, i.e. purchases by the same investor following advice that do not match any of the recommendations. Both portfolios use fixed initial weights. For both of these portfolios, I calculate market beta, annualized volatility, relative Sharpe-ratio loss, and maximum drawdown using a common 60-month return history. Table F.1 reports how these portfolios differ.

The portfolios differ in terms of the fund share, fees, and historical risk. The relative Sharpe-ratio loss is 10.7 percentage points lower for recommended portfolios, falling from 28.1 to 17.4 percent, a 38.1 percent reduction. They retain 82.6 percent of market Sharpe, compared with 71.9 percent for the alternative purchase portfolios. The annualized volatility is 11.27 percentage points lower and the market beta is 0.231 lower; the maximum peak-to-trough decline is 10.35 percentage points smaller. The purchased-fund expense ratios are 9.09 basis points lower for advice-following purchases than for other post-advice purchases (Table F.2).

Figure 4 in addition reports how both implemented recommendations and other investments line up

with pre-advice portfolio characteristics. It shows that market betas tend to be lower for advice-following purchases than for “non-compliant” purchases. Annualized volatility is also lower. Underlying-asset HHI is lower for advice-following purchases than for other purchases. The ETF/fund share of purchases is higher for advice-following purchases. Relative Sharpe-ratio loss is lower for advised purchase portfolios than for other purchase portfolios. Comparisons are made within investor and response window. Figure 4 also addresses concerns about the sycophancy of generative AI advice in financial contexts: recommendations differ from non-recommended investments along several dimensions, and similarly also differ from existing portfolio holdings.

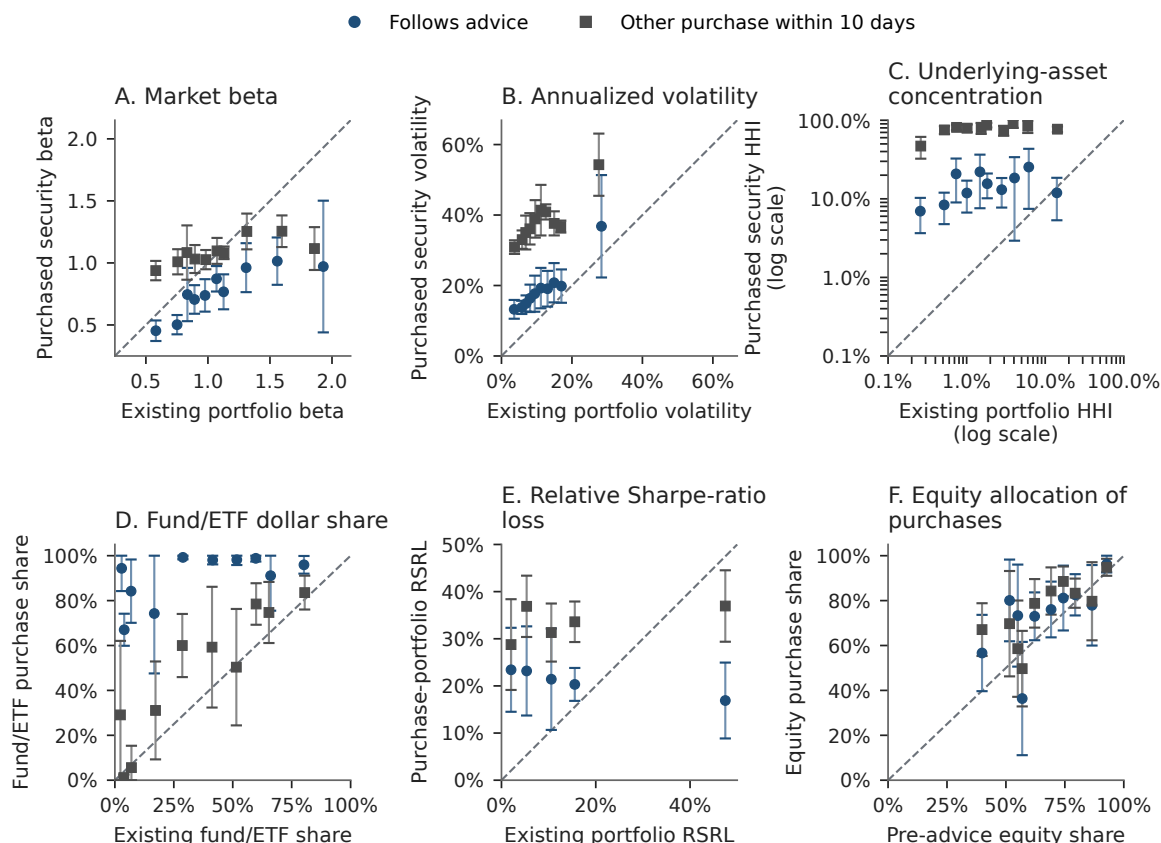


Figure 4. Advice-following trades and existing portfolio characteristics

Notes: This figure plots equal-event means within five groups of the corresponding pre-advice characteristic. The error bars indicate 95 percent confidence intervals, with standard errors clustered by individual. Relative Sharpe-ratio loss is computed following Calvet et al. (2007).

Table F.2 reports both purchase composition and the risk and fee characteristics of purchased securities. Transactions that match AI advice, relative to other post-advice purchases, are 36.34 percentage points more likely to be diversified funds, 40.86 percentage points less likely to be individual stocks, 10.66 percentage points more likely to be broad passive equity funds, and have expense ratios that are 9.09 basis points lower.

In addition to immediate adherence to advice, I examine whether advice accumulates in retail investors' portfolios. Even if investors implement generative AI advice, adherence could be temporary and shifts in allocations may not be economically meaningful. Figure 5 plots the share of investors' final observed noncash

portfolios invested in AI-advised tickers against time since first advice and cumulative interactions. For this figure, I restrict to buy advice that introduces a ticker that is neither mentioned in the prompt nor recorded in the investor’s transaction history before the recommendation. Panel A plots the advised share against cumulative interactions; Panel B plots it against days since first advice. At the final observed portfolio date, 30.6 percent of investors hold at least one ticker introduced to them by gen AI advice.

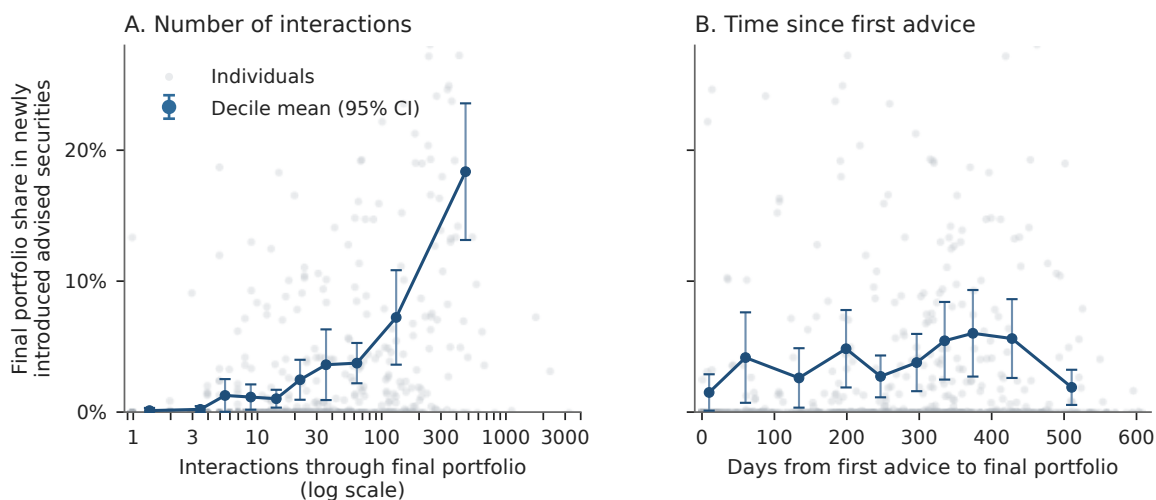


Figure 5. Accumulation of newly introduced advised investments in investor portfolios

Notes: This figure shows the relationship between advice exposure and the advised share of the final observed positive noncash portfolio. I define advised securities as exact-ticker buy recommendations absent from the individual’s prompt and prior transaction history. The horizontal axis measures cumulative classified answer interactions on a logarithmic scale in Panel A and days since first advice in Panel B. Gray points represent individuals; blue points report means within deciles, with 95 percent confidence intervals.

Figure F.1 reports the same accumulation statistics but instead for any ticker recommended by AI. For heavier users, a higher share of their portfolio is “advised” in the sense that they have received AI ticker-level advice. Taken together, these figures provide descriptive ticker-level evidence that investors hold securities introduced by generative AI. The evidence in this section complements the main DID estimates in Section 5 by linking transactions and holdings to recommendations received in specific conversations.

6.2 Who Takes Advice, and Why?

Implementation varies substantially across users and over time. Individual fixed effects explain 26.4 percent of the variation in answer-level implementation; adding answer-date fixed effects raises this share to 33.9 percent. A possible explanation for this variation is variation in how the advice is given. Overall, the high implementation rates of advice contrast with those in comparable human-advice settings: Bhattacharya et al. (2012) report that only about 5 percent of investors offered free, unbiased advice obtain it, and those who do follow little of it. One dimension along which generative AI advice may differ from human advice is the degree of personalization. Foerster et al. (2017) find that Canadian financial advisers provide limited customization of clients’ portfolios. The additional variation explained by answer-date fixed effects may point towards market conditions influencing when individuals are willing to implement advice. I also examine

whether recent portfolio returns are associated with implementation.

To explore these hypotheses, I regress implementation on personalization. Table 4 reports estimates from this regression. Including answer-date purchases, the implementation rate is 43.06 percent. Personalization is measured in two different ways: first, using a local LLM directly to assign a personalization score. Second, this measure is compared to one constructed from summing up personalization details across the following four categories: portfolio state, life details, preferences and constraints, and goals. The correlation between these two measures is 0.78.

Panel A reports results for the LLM-generated measure, and Panel B for the number of personalization details. One standard deviation increase in personalization details (4.92 details) is associated with a 5.14-percentage-point higher implementation rate. This estimate is robust to adding individual and date fixed effects: with two-way fixed effects, the association is 3.40 percentage points. For the other measure, the estimates are 5.64 and 3.23 percentage points, respectively.

Table 4. Personalization measures and advice implementation

	Advice implementation within ten trading days			
	(1)	(2)	(3)	(4)
<i>Panel A. LLM personalization score</i>				
LLM personalization score	2.71*** (0.66)	1.71*** (0.38)	1.57*** (0.34)	1.55*** (0.29)
One-standard-deviation increase (pp)	5.64*** (1.38)	3.56*** (0.79)	3.26*** (0.70)	3.23*** (0.61)
R^2	0.013	0.256	0.140	0.332
<i>Panel B. Number of personalization details</i>				
Number of personalization details	1.04*** (0.35)	0.67*** (0.17)	0.65** (0.26)	0.69*** (0.13)
One-standard-deviation increase (pp)	5.14*** (1.71)	3.30*** (0.81)	3.19** (1.26)	3.40*** (0.62)
R^2	0.011	0.256	0.140	0.333
Outcome mean (%)	43.06	43.06	43.06	43.06
Individual fixed effects	No	Yes	No	Yes
Answer-date fixed effects	No	No	Yes	Yes

Notes: This table reports separate linear probability models of implementation on personalization. The dependent variable is an indicator equal to one if at least one recommended security is purchased on the answer date or during trading days 1–10 afterward. I measure personalization in two ways: the LLM’s 1–10 score in Panel A and the number of recorded details about portfolio state, life details, preferences and constraints, and goals within each answer in Panel B. The correlation between the two measures is 0.78. Coefficients are reported in percentage points per additional score point or detail, and for a one-standard-deviation increase in the same measure. Each specification includes only that measure and the indicated fixed effects. Standard errors are clustered at the individual level and reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 6 shows a positive relationship between personalization and implementation using purchases during trading days 1–10 after the answer. Panel A plots raw implementation rates across five personalization-score bands. Panel B shows that the positive relationship remains after controlling for individual and exact answer-date fixed effects. A one-point increase in the personalization score is associated with a 1.45-percentage-point increase in implementation (standard error 0.28). Thus, the relationship is not explained solely by persistent differences across investors or factors common to answers received on the same date.

Table F.4 reports the impact of different personalization categories. A unique feature of the portfolio-related personalization is that it is driven by the portfolio context the AI assistant has access to. It is thus not subject to confounding features of the investor-written prompt. Some portfolio details appear in 70 percent of answers that provide ticker-level advice.

Panel A of Table F.4 shows that implementation is higher when advice uses account-location, cash, and transaction or gains-and-loss details. These coefficients remain statistically significant after controlling for individual and answer-date fixed effects. In this specification, these details are associated with 3.17, 4.01, and 3.54 percentage points higher implementation, respectively. These results are consistent with the idea that practical account information makes recommendations easier to evaluate and implement.

Panel B of Table F.4 shows that concentration mentions are associated with a larger increase in the fund share of new purchases. Among first broad-fund-advice events, a concentration mention is associated with a 5.11-percentage-point larger increase in the fund share of positive purchase dollars over the following four weeks (standard error 2.54) in the preferred receiver-only, trait-adjusted specification. The corresponding estimate in the full stacked specification is 5.43 percentage points. These results are consistent with the idea that drawing attention to portfolio concentration encourages purchases of diversified funds.

The results in Table F.5 show that investors are more likely to implement advice following weak short-run portfolio performance. In column (2), a 1-percentage-point lower five-day portfolio return relative to VTI is associated with a 1.93-percentage-point higher implementation rate (standard error 0.66). This pattern is not robust across return horizons and portfolio definitions (Tables F.5 and F.6).

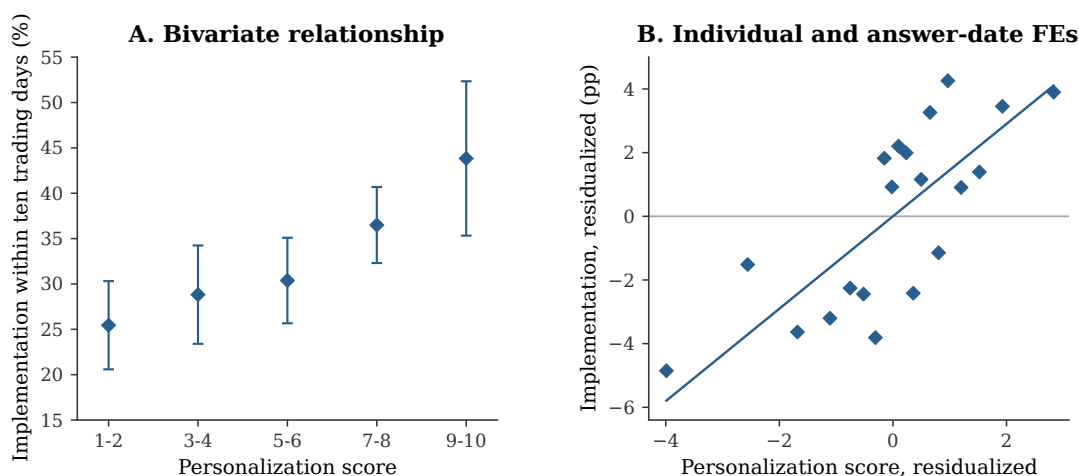


Figure 6. Personalization and implementation within ten trading days

6.3 Financial Vocabulary in Subsequent Prompts

I find suggestive evidence that investors learn from interacting with generative AI: financial terms used in AI answers are more likely to appear in investors' subsequent prompts, including in later conversations. Whether AI helps individuals acquire knowledge, beyond providing answers to their immediate questions, is a central issue in the literature on AI and learning (Brynjolfsson et al., 2025; Bastani et al., 2025). I examine one potential indication of learning: whether individuals subsequently use financial concepts introduced by generative AI in their own questions. To do so, I estimate the effect of AI introducing a financial concept on subsequent prompts by investors.

I estimate the following regression at the individual–concept–conversation level:

$$Y_{ic,j+1} = \beta \text{AIIntro}_{icj} + \alpha_i + \tau_{w(j)} + \varepsilon_{icj}, \quad (3)$$

The outcome variable, $Y_{ic,j+1}$, equals one if individual i uses concept c in the first prompt of conversation $j + 1$. The sample requires the next conversation to occur within 90 days. My main variable of interest is AIIntro_{icj} , an indicator that equals one if generative AI uses the concept in conversation j before the individual has used it in that or any earlier conversation. I include individual fixed effects, α_i , and calendar-week fixed effects, $\tau_{w(j)}$, where $w(j)$ denotes the week of conversation j . Standard errors are clustered by individual.

For each individual and concept, the sample retains conversations until the first observed use by either the individual or AI. I group different forms of the same word under one concept. The main estimates pool four concepts: fund, allocation, diversification, and index. I add volatility, concentration, and risk for the concept-specific examples in Panel B of Table 5.

Table 5. AI introduction and subsequent use of financial concepts

	Individual uses concept in first prompt of next conversation		
	(1)	(2)	(3)
<i>Panel A. Pooled estimate</i>			
AI introduces concept	3.19*** (0.54)	2.07*** (0.60)	1.98*** (0.61)
Outcome mean	4.67	4.67	4.67
Total R^2	0.005	0.196	0.204
Individual fixed effects	No	Yes	Yes
Calendar-week fixed effects	No	No	Yes
<i>Panel B. Examples of concept-specific estimates</i>			
Concept	Estimate	Concept	Estimate
Fund	4.43*** (0.93)	Allocation	8.16*** (1.17)
Diversification	4.52*** (0.98)	Volatility	-3.36*** (0.36)
Concentration	-2.46*** (0.49)	Risk	11.21*** (1.34)

Notes: The unit of observation is an individual–concept–conversation. The sample includes conversations up to the first observed use of the concept by either the individual or AI. The dependent variable is an indicator equal to one if the individual uses the concept in the first prompt of the next conversation, within 90 days. The main regressor is an indicator equal to one if generative AI first introduces the concept. Panel A pools fund, allocation, diversification, and index. Panel B adds volatility, concentration, and risk and reports six examples from the seven-concept model with individual and calendar-week fixed effects. Different forms of the same word are grouped together. Coefficients and means are expressed in percentage points. Column (1) includes no fixed effects; column (2) adds individual fixed effects; column (3) adds calendar-week fixed effects. Standard errors are clustered at the individual level and reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A of Table 5 shows a positive relationship between AI introduction and subsequent use of financial concepts. After controlling for individual and calendar-week fixed effects, the estimated coefficient is 1.98 percentage points (standard error 0.61), compared with an outcome mean of 4.67 percent. The coefficients decline with the inclusion of fixed effects, but remain positive and statistically significant.

Panel B reports the results separately for individual concepts. The coefficients on fund, allocation, diversification, and risk are positive and statistically significant. In contrast, the coefficients on volatility and concentration are negative and statistically significant.

Table 6. Algorithmically selected financial terms and subsequent individual use

<i>Panel A. Prevalence-based term selection</i>			
Term	AI answers (%)	Individual prompts (%)	Ratio
Exposure	37.6	6.0	6.3×
Allocation	31.3	6.1	5.1×
IRA	30.2	5.0	6.1×
Yield	21.4	1.6	13.1×
Volatility	20.6	0.6	36.4×
Diversification	17.7	2.5	7.0×
Liquidity	13.5	0.6	24.5×
Roth	13.2	0.9	15.6×
<i>Panel B. Subsequent use in the next conversation</i>			
	(1)	(2)	(3)
AI introduces term	2.94*** (0.31)	2.52*** (0.33)	2.50*** (0.33)
Outcome mean	2.08	2.08	2.08
Total R^2	0.006	0.069	0.071
Individual fixed effects	No	Yes	Yes
Calendar-week fixed effects	No	No	Yes

Notes: Panel A reports the prevalence of selected financial terms that meet the screen. I retain exact words or phrases appearing in at least 10 percent of AI answers and at least three times as often in answers as in individual prompts. Selection does not use subsequent individual language. I measure prevalence as the share of individual-written prompt–answer events containing the term. Panel B repeats the risk-set construction and outcome in Table 5 for the eight terms shown in Panel A. Coefficients and means are expressed in percentage points. Column (1) includes no fixed effects; column (2) adds individual fixed effects; column (3) adds calendar-week fixed effects. Standard errors are clustered at the individual level and reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6 approaches the selection of relevant terms systematically. It retains only terms that appear at least three times as often in AI answers as in prompts and that appear in at least 10 percent of AI answers. Appendix Table F.7 reports the retained financial terms that meet these criteria. Panel A reports the prevalence of selected financial terms in AI answers and prompts.

7 Conclusion

Generative AI has become one of the most rapidly adopted technologies. This rapid adoption extends to financial advice, with a recent Investing.com survey reporting that nearly two-thirds of U.S. retail investors have used generative AI tools to assist with investment decisions (Investing.com, 2026).

This paper offers the first assessment, to my knowledge, of how generative AI financial advice affects observed investment decisions and portfolio allocations. Using linked records of investors' questions, AI answers, security-level recommendations, transactions, and holdings, I find that generative AI predominantly recommends funds in response to open-ended requests for investment ideas. After investors begin using generative AI, their purchases shift toward broad passive index funds, and this shift subsequently appears

in portfolio holdings. The securities they purchase carry less idiosyncratic risk, and the funds they select charge lower fees. These portfolio changes address problems widely documented in household finance: retail investors often hold under-diversified portfolios and bear uncompensated idiosyncratic risk (Calvet et al., 2007; Badarinza et al., 2016), and they frequently select investment products with unnecessarily high fees or otherwise undesirable fund characteristics (Choi et al., 2010; Bailey et al., 2011).

This shift in investment behavior is driven by AI advice consistently directing investors toward greater diversification through low-cost index investing, as prescribed by standard financial theory (Markowitz, 1952; Campbell and Viceira, 2002; Campbell, 2006). I find high rates of advice implementation. This is notable given extensive evidence that changing household financial behavior is difficult (Madrian and Shea, 2001; Bhattacharya et al., 2012; Fernandes et al., 2014; Andersen et al., 2020). I find suggestive evidence that one feature of AI advice that may distinguish it from more generic sources of financial guidance is its ability to identify risks in an investor's existing portfolio. By calling attention to portfolio concentration, generative AI may change investors' beliefs about the riskiness of their holdings and make them more willing to direct subsequent purchases toward diversified funds.

Overall, my results show that generative AI advice directs investors toward diversified, low-cost index funds and that investors implement a substantial share of this guidance. The resulting changes appear in both purchases and portfolio holdings, reducing idiosyncratic risk and fund expenses. Future research using larger samples and longer account histories can determine whether these individual-level effects persist and aggregate into meaningful changes in fund flows, security demand, trading, and prices.

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Appendix

A Data and Advice

Table A.1. Summary statistics: Mezzi and the 2022 Survey of Consumer Finances

Variable	Mezzi			2022 SCF		
	Mean	SD	Median	Mean	SD	Median
<i>Panel A. Sample and Demographic Characteristics</i>						
Equity-owning sample, <i>N</i>	1,854	–	–	2,966	–	–
Age (years)	52.9	12.9	55.0	52.1	16.8	52.0
Married (%)	79.6			67.3		
Any children/dependents (%)	54.0			39.7		
Number of children/dependents	0.91	0.99	1.00	0.71	1.05	0.00
<i>Panel B. Portfolio Characteristics</i>						
Hold > 1 direct stock (%)	75.0			28.2		
Portfolio value (\$000)	1,908.9	2,925.8	555.7	845.3	5,986.9	155.0
Retirement share of investable wealth (%)	33.9	36.4	19.3	71.6	37.3	96.8
<i>Portfolio composition</i>						
Direct share of total equity (%)	44.9	32.6	41.0	45.4	38.5	35.9
Number of direct stocks	39.3	62.7	13.0	10.4	20.0	4.0
Total-equity share of portfolio (%)	70.4	23.4	75.3	46.1	29.3	44.0
Indirect-equity share of portfolio (%)	42.5	25.5	43.4	41.7	28.2	37.8
Direct-equity share of portfolio (%)	33.6	29.4	25.3	20.8	24.1	10.9

Notes: This table reports summary statistics for Mezzi investors and equity holders in the 2022 Survey of Consumer Finances (SCF). The SCF sample consists of 2,966 distinct equity-owning observations; the five implicates are not separate observations. For Mezzi, the statistics are computed using each investor's latest positive connected portfolio. For binary measures, the Mean column reports the percentage satisfying the condition; SD and Median are left blank. The portfolio measures are defined following Polkovnichenko (2005). The sample is restricted to direct-stock owners for direct-share and stock-count statistics, and to holders of the corresponding equity type for portfolio shares. I measure SCF shares relative to financial assets as defined in that paper, and Mezzi shares relative to observed connected portfolio value. Indirect equity includes funds, retirement accounts, annuities, trusts, and managed accounts. Retirement share is the ratio of retirement-account wealth to investable wealth.

Table A.2. Monthly portfolio turnover

Sample	Account coverage	Mean (%)	Median (%)
Mezzi	All linked accounts	8.39	2.09
Mezzi	Largest account only	7.50	1.55
Barber–Odean (2000)	One brokerage	6.49	2.67
Giglio et al. (2021)	Vanguard only	2.30	0.91

Table A.3. Investor Buy Intent and Gen AI Buy Advice

Share of messages (%)

	Investors' prompts (buy intent)	Gen AI answers (buy advice)
Stocks	1.9	4.6
With an individual-stock ticker	1.9	4.4
ETFs/funds	3.5	18.0
With an ETF/fund ticker	1.7	15.7

Notes: This table compares investors' prompts and the corresponding generative AI answers. The measures are interest in buying for prompts and buy recommendations for answers. I exclude tickers mentioned only as existing holdings and assign equal weight to each investor.

[illegible][illegible]

Notes: This figure summarizes the words used in prompts and their corresponding AI answers. The sample excludes exact duplicate prompts, and each prompt–answer pair receives equal weight. I measure word frequency as the number of prompts or answers containing the word. Word size is proportional to frequency, with each panel normalized separately. I exclude common stopwords, conversational instructions, redaction markers, and platform artifacts, and combine plural variants of common financial terms.

B Outcome Definitions

B.1 Portfolio Construction

I reconstruct daily portfolios from transactions, transfers, reinvestments, and dividend receipts. For positions that predate the available transaction history, the number of shares is calculated as the recorded dividend or fund distribution divided by the corresponding CRSP distribution per share. Purchases, sales, reinvestments, and split adjustments update each position.

The market value of each position is calculated as the shares held times the daily closing price per share. I calculate total portfolio value by summing positions across linked accounts. I compare the reconstructed holdings with periodic holdings snapshots. From these data, I construct daily measures of market value, asset-class composition, concentration, and returns.

B.2 Variable Definitions

I define the following portfolio shares:

$$\text{Equity share} = \frac{\text{Equity holdings}}{\text{Total portfolio value}} \times 100,$$

$$\text{Passive-fund portfolio share} = \frac{\text{Passive index fund holdings}}{\text{Total portfolio value}} \times 100,$$

$$\text{Individual-stock portfolio share} = \frac{\text{Individual-stock holdings}}{\text{Total portfolio value}} \times 100.$$

Total portfolio value is the value of all holdings across linked accounts. For the regression outcomes, I use the following purchase and month-end equity shares:

$$\text{Passive-fund purchase share} = \frac{\text{Passive index fund purchases}}{\text{Purchases}} \times 100,$$

$$\text{Individual-stock purchase share} = \frac{\text{Individual-stock purchases}}{\text{Purchases}} \times 100,$$

$$\text{Passive-fund holdings share} = \frac{\text{Passive index fund holdings}}{\text{Equity holdings}} \times 100.$$

Purchases and holdings are measured in dollars.

Table B.1. Principal outcomes

Outcome	Definition
Direct-index purchase share	The value of CRSP D direct-index fund purchases divided by the total value of purchases. Leveraged, inverse, and alternative products are excluded.
Net fund–stock tilt	Net purchases of ETFs/funds plus net sales of individual stocks, expressed as a percentage of total absolute noncash trading.
Fund purchase fee	Purchase-dollar-weighted expense ratio (basis points; fee-covered purchases).
Response implementation	1 if any exact recommendation in the answer is implemented; 0 otherwise.
Recommendation-level implementation	1 if the specific security recommendation is implemented; 0 otherwise.
Unadvised passive purchase	Passive-fund purchase without a recent same-ticker recommendation.

C Changes in Recommendation Actionability

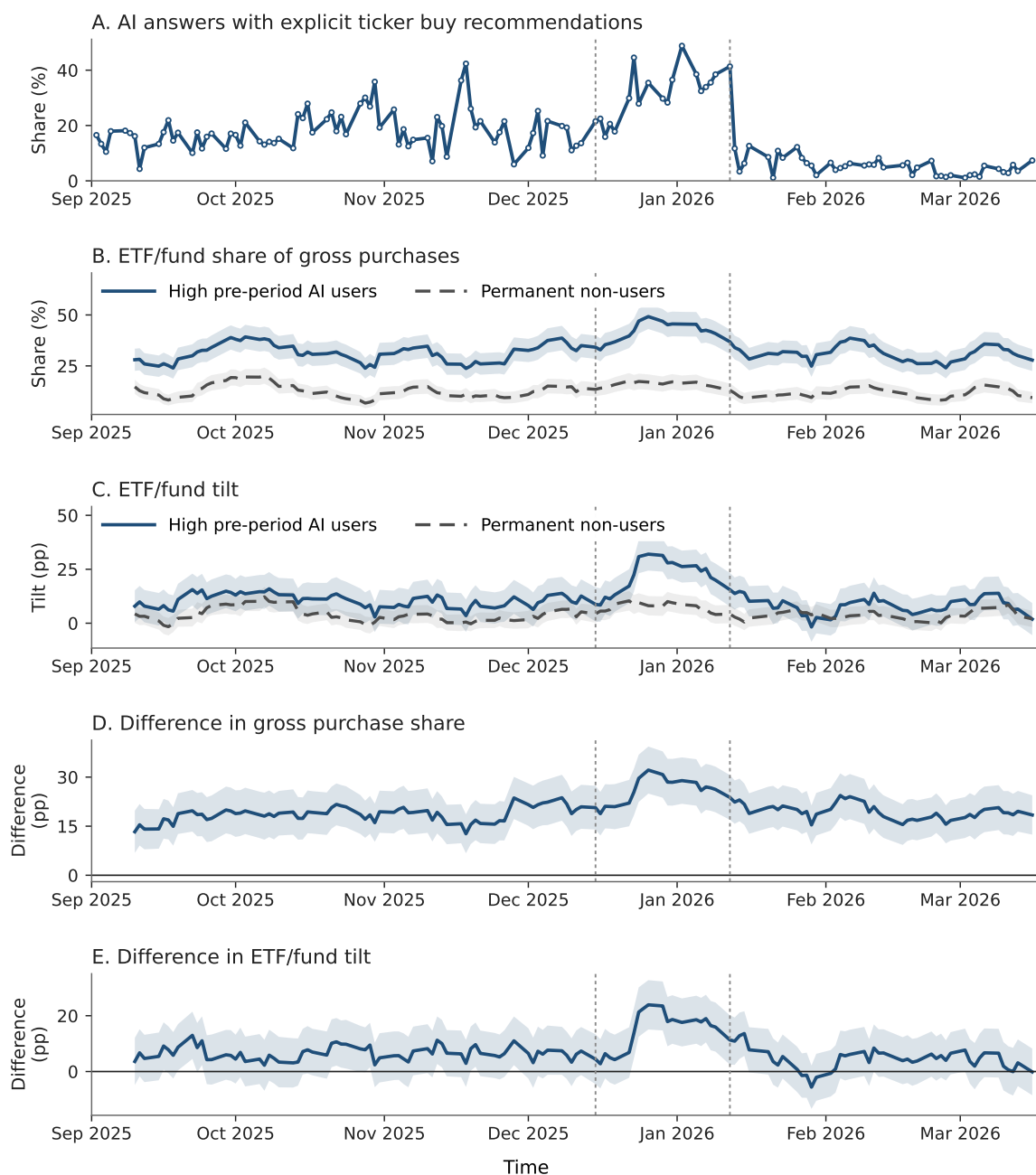


Figure C.1. Recommendation actionability and ETF/fund purchases

Notes: Panel A shows the daily share of AI answers containing an explicit security-level buy or add recommendation. Panels B and C compare the highest pre-period AI-activity quartile with permanent non-users. The measures are the ETF/fund share of gross noncash purchase dollars (B) and fund purchases plus stock sales minus fund sales and stock purchases, divided by gross noncash trading (C). Panels D and E report the differences between these groups for B and C. The shaded areas in B–E indicate pointwise 95 percent confidence intervals. Standard errors use cross-investor variation in daily means. Groups are defined using AI activity from November 12, 2024 to August 31, 2025 and held fixed. The purchase measures are computed over complete trailing-seven-session windows continuing across both platform-change dates; windows without relevant trades are zero.

Table C.1. System-prompt changes and gross ETF/fund purchase shares

	High minus Pre (1)	Low minus High (2)
Adjacent-phase DID	8.13*** (1.95)	-7.67*** (2.15)
Observations	15,316	15,316
Calendar weeks	28	28
R^2	0.484	0.484
Individual FE	Yes	Yes
Calendar-week FE	Yes	Yes

Notes: This table reports regressions of the ETF/fund share of gross purchases on system-prompt changes. The unit of observation is an individual-calendar week. The dependent variable is defined as 100 times ETF/fund gross purchase dollars divided by all positive-dollar noncash gross purchase dollars; weeks without purchases are zero. The treated and comparison groups consist of the highest pre-period AI-activity quartile and permanent non-users, respectively. Columns (1) and (2) report High minus Pre and Low minus High, respectively, as linear contrasts from the same regression. All specifications include individual and calendar-week fixed effects. Standard errors are double-clustered by individual and calendar week and reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Additional Results and Robustness Checks

Table D.1. Effects of AI Adoption on Purchase Characteristics

	Full purchase portfolio			Stock-only purchases	Fund expense ratio
	U.S. market	World market	Welch BSWA (Welch, 2022)	U.S. market	Basis points
	(1)	(2)	(3)	(4)	(5)
Post-adoption effect	−0.146*** (0.042)	−0.116*** (0.031)	−0.122*** (0.037)	0.011 (0.024)	−3.57** (1.41)
DV mean	2.67	2.77	2.27	3.57	29.05
Individual fixed effects	Yes	Yes	Yes	Yes	Yes
Calendar-month fixed effects	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable in columns (1)–(4) is log idiosyncratic volatility under the indicated market model. Columns (1)–(3) use the combined monthly purchase portfolio, while column (4) is restricted to individual-stock purchases. In column (5), the dependent variable is the purchase-dollar-weighted expense ratio of funds bought that month. Standard errors are double-clustered by individual and calendar month and reported in parentheses. Separately, CCS return loss (Calvet, Campbell, and Sodini, 2007) falls by 39.9 percent for the combined purchase portfolio. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.1 Purchase-Risk and Fund-Fee Distributions

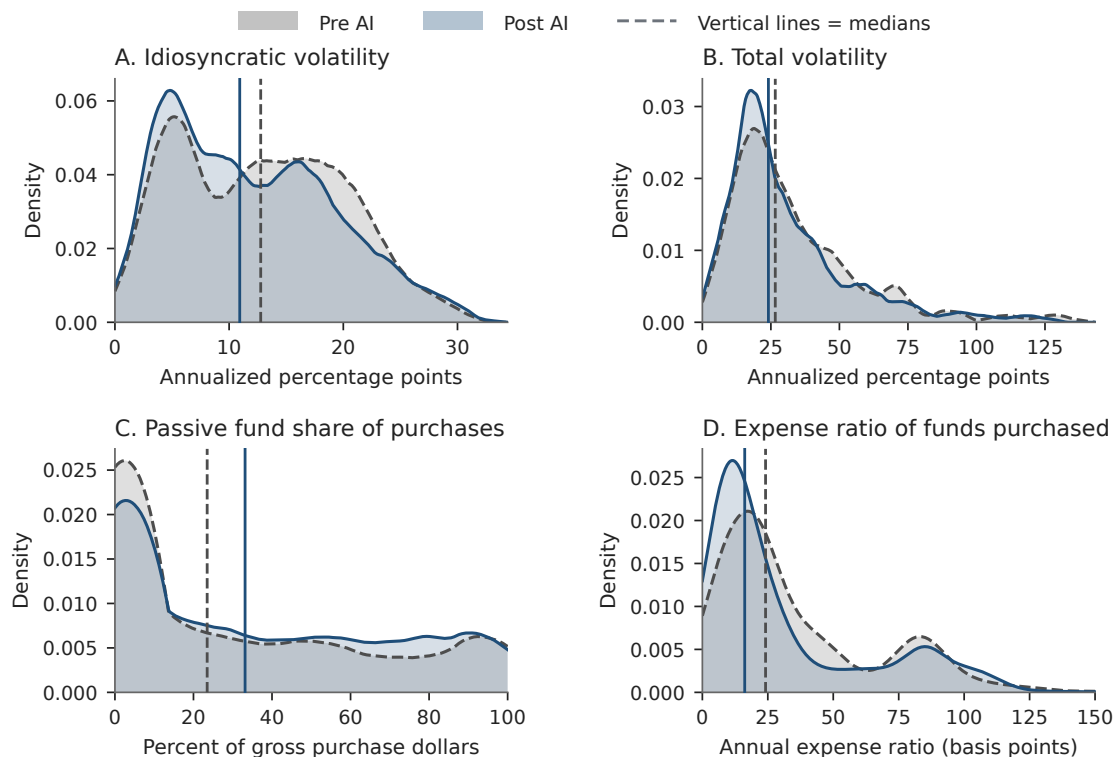


Figure D.1. Purchase-portfolio distributions before and after AI adoption

Notes: This figure shows the distributions of purchase-portfolio characteristics before and after AI adoption. In Panels A–C, the sample consists of available adopter observations in event months -1 and $+1$, with equal individual weights within each period; individuals need not appear in both periods. Panels A and B measure purchase-dollar-weighted annualized volatility. Panel C measures strict passive-index ETF and mutual-fund purchases as a share of purchases. Panel D uses monthly expense ratios in months -9 to -1 and $+1$ to $+9$, weighted by fee-covered fund-purchase dollars. Month 0 is excluded. The curves show kernel densities, and the vertical lines indicate full-sample medians. For plotting only, Panels A–C exclude observations at or above the pooled 99th percentile.

Panel	Pre median	Post median	Change	<i>p</i> -value
A. Idiosyncratic volatility (%)	12.77	10.93	-1.84	0.004
B. Total volatility (%)	26.60	24.10	-2.50	0.005
C. Passive-fund purchase share (%)	23.49	33.14	+9.65	0.007
D. Fund expense ratio (bp)	24.14	16.18	-7.96	0.031

Changes equal post minus pre medians, in percentage points for Panels A–C and basis points for Panel D. Two-sided *p*-values test equality of the pre/post distributions. Tests use the full unsmoothed samples and do not adjust for calendar-time effects.

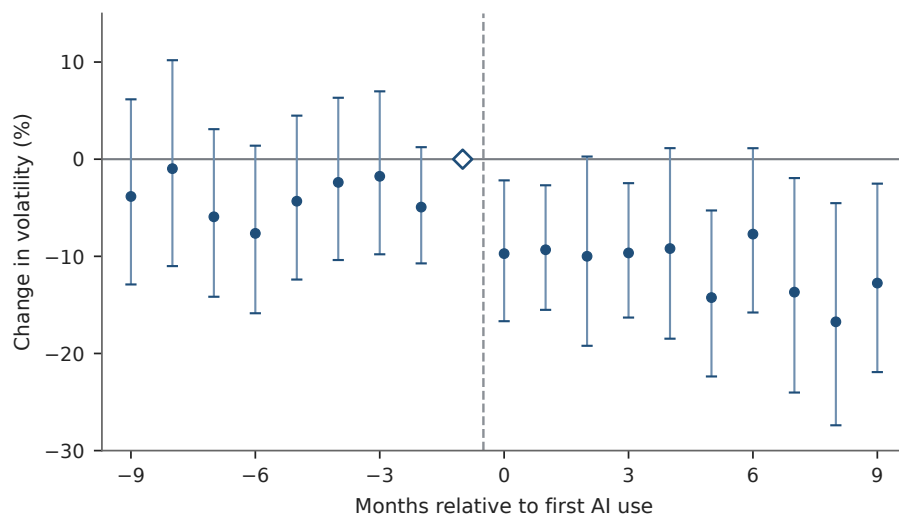


Figure D.2. Idiosyncratic volatility of purchases following AI adoption

Notes: This figure plots the estimated coefficients and 95 percent confidence intervals from stacked event-study regressions. The dependent variable is the log of purchase-dollar-weighted annualized residual volatility of purchased securities, estimated using an MSCI World raw-return OLS market model. The coefficients are transformed to exact percentage effects. The omitted period is month -1 . All regressions include stack-by-individual and stack-by-calendar-month fixed effects. Standard errors are double-clustered by individual and calendar month.

Table D.2. Concentration and volatility of individual-stock purchases

	Stock-purchase HHI (1)	Log total volatility (2)	Log idiosyncratic volatility (3)
AI adoption	2.28 (1.94)	0.48 (2.20)	1.11 (2.36)
R^2	0.58	0.59	0.59
Calendar-month FE	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes
Outcome mean	44.25	40.36	33.30

Notes: This table reports Sun–Abraham interaction-weighted estimates averaged over event months 1–9. The sample is restricted to months with total purchases of at least 0.5 percent of the opening portfolio and a positive individual-stock purchase. The dependent variables in columns (1)–(3) are 100 times the sum of squared stock-purchase-dollar weights, log total volatility, and log idiosyncratic volatility, respectively. I compute volatility using the stocks purchased and a common November 2019–October 2024 return history. Column (3) uses CRSP value-weighted market-model residual volatility. Return coverage must reach 80 percent of stock purchase dollars. Log coefficients are multiplied by 100 and approximate percentage changes; outcome means are in levels. All specifications include individual and calendar-month fixed effects. The omitted period is month -1 . Standard errors are double-clustered by individual and calendar month and reported in parentheses. Significance levels: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

D.2 Fund-Purchase Fees

Table D.3. Robustness of fund-purchase fee estimates

	Weighted stacked DID (1)	Two-way fixed effects (2)	Cohort-specific event study (3)	Group-time ATT (4)	Any fund purchase (5)
AI adoption	-2.18 (2.29)	-3.43*** (1.30)	-3.57** (1.41)	-3.73 (2.38)	-2.20** (0.91)

Notes: The dependent variable is the purchase-dollar-weighted annual expense ratio of fee-covered ETF and mutual-fund purchases, measured in basis points. Columns (1)–(4) restrict the sample to the preferred activity sample. Column (5) includes all fee-covered fund purchases and uses the cohort-specific event-study estimator. The reported effects are averaged over event months 1–9, relative to month –1. In column (1), the event-specific datasets are stacked together. Standard errors in parentheses are double-clustered by individual and calendar month in columns (1)–(3) and (5). Column (4) uses standard errors clustered by individual. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.3 Heterogeneity by Pre-AI Diversified-Fund Share

Table D.4. Monthly purchase portfolio effects by pre-AI diversified-fund share

	Sun–Abraham	Stacked DID
Quintile 1	7.45*** (1.81)	10.20*** (2.55)
Quintile 2	14.35*** (2.53)	13.39*** (4.21)
Quintile 3	6.82*** (2.49)	6.03 (4.39)
Quintile 4	3.99 (2.68)	4.34 (4.67)
Quintile 5	1.04 (1.98)	1.76 (4.75)
Individual fixed effects	Yes	Yes
Date fixed effects	Yes	Yes

Notes: This table reports estimates separately by pre-AI diversified-fund share quintile. The dependent variable is the direct-index fund share of purchases. The sample is restricted to months satisfying the 0.5 percent activity requirement. Treated individuals are assigned once to quintiles, with higher quintiles indicating larger pre-AI fund shares. The Sun–Abraham estimates average event months 1–9 relative to month -1 . The comparison group for stacked DID consists of fixed individuals untreated through $g + 12$. Standard errors are double-clustered by individual and calendar month and reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.4 Robustness

Table D.5. Passive-Fund Shares across Difference-in-Differences Estimators

	Point estimate	Standard error	Lower bound	Upper bound
<i>Panel A. Monthly Purchase Portfolio</i>				
Sun–Abraham	7.60***	1.55	4.55	10.65
Stacked DID	7.73***	2.20	3.42	12.05
Callaway–Sant’Anna	6.11**	2.81	0.61	11.61
Borusyak–Jaravel–Spiess	5.32***	1.93	1.53	9.12
de Chaisemartin–D’Haultfoeuille	6.03***	2.18	1.76	10.29
TWFE–OLS	4.98***	1.61	1.82	8.13
<i>Panel B. Selling-inclusive net share, 0.5 percent activity requirement</i>				
Sun–Abraham	7.53***	2.23	3.16	11.90
Stacked DID	11.18**	4.34	2.67	19.69
Callaway–Sant’Anna	6.91	4.25	–1.42	15.23
TWFE–OLS	6.08**	2.78	0.64	11.52
<i>Panel C. Total Portfolio</i>				
Sun–Abraham	2.73***	0.68	1.41	4.06
Stacked DID	1.38	0.91	–0.41	3.17
Callaway–Sant’Anna	2.40**	1.13	0.18	4.61
TWFE–OLS	1.57*	0.86	–0.12	3.26

Notes: This table reports estimates using alternative difference-in-differences estimators. The dependent variables use the direct-index definition, and coefficients are expressed in percentage points. The sample for the Monthly Purchase Portfolio and Panel B is restricted to months with purchases of at least 0.5 percent of the opening portfolio. The headline net-share column includes every trading month. Standard errors are double-clustered by individual and calendar month, except for Callaway–Sant’Anna, BJS, and dCDH, which use household-clustered inference. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

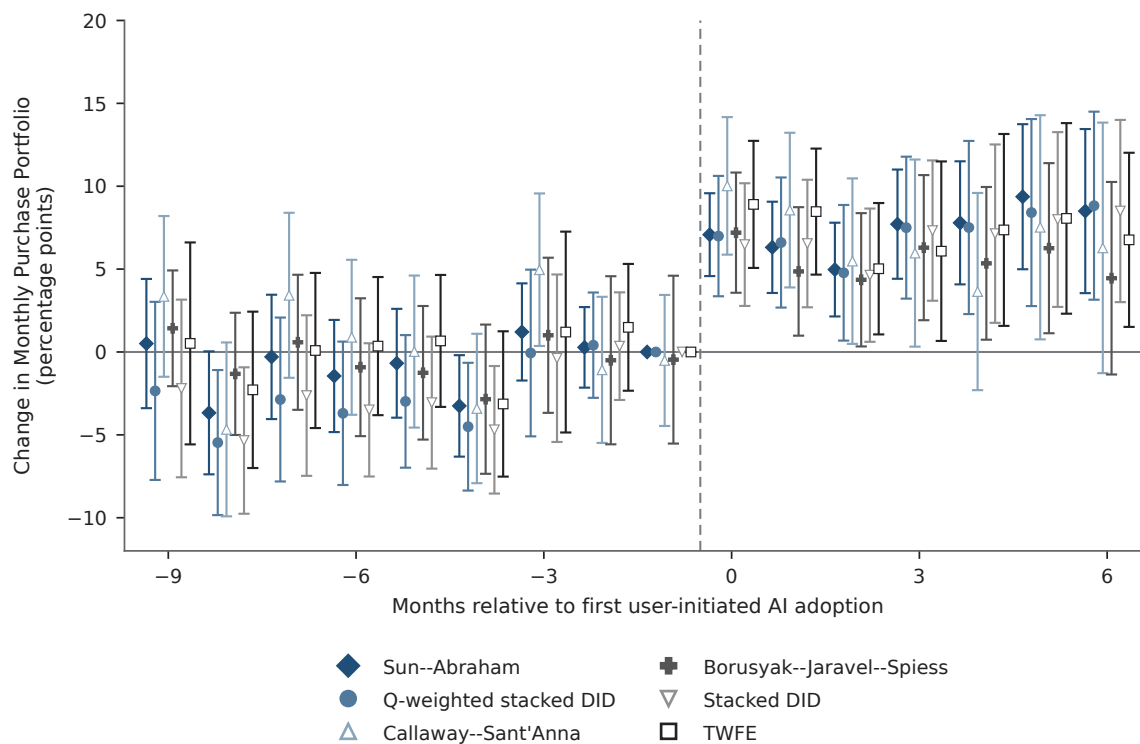


Figure D.3. Monthly Purchase Portfolio across Difference-in-Differences Estimators

Notes: This figure plots the estimated coefficients and 95 percent confidence intervals for the direct-index fund share of the Monthly Purchase Portfolio. The stacked regressions use fixed cohort-specific comparison samples. The omitted period is month -1, except that BJS pre-period coefficients are placebo estimates.

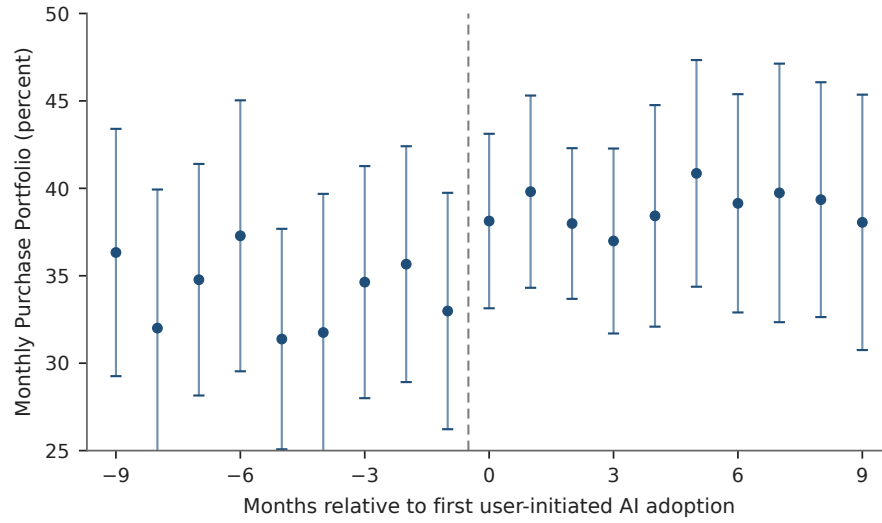


Figure D.4. Unadjusted Monthly Purchase Portfolio among AI Adopters

Notes: This figure shows unadjusted mean direct-index fund shares of purchases among adopters. The horizontal axis measures months relative to first user-initiated AI adoption. The error bars indicate 95 percent confidence intervals. The sample excludes zero-purchase months, for which the purchase share is undefined.

Table D.6. Alternative Passive-Fund Definitions

Specification	Point estimate	Standard error
<i>Panel A. Monthly Purchase Portfolio</i>		
Direct-index funds	7.60***	1.55
CRSP B/D/E index funds	6.38***	1.46
<i>Panel B. Month-End Portfolio</i>		
Direct-index funds	2.73***	0.68
CRSP B/D/E index funds	2.50***	0.68

Notes: This table reports Sun–Abraham estimates using alternative passive-fund definitions. The dependent variable is the passive-fund share of the monthly purchase portfolio in Panel A and the passive-fund share of month-end equity holdings in Panel B. Panel A restricts the sample to months with purchases of at least 0.5 percent of the opening portfolio. All specifications include individual and calendar-month fixed effects. Standard errors are double-clustered by individual and calendar month and reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

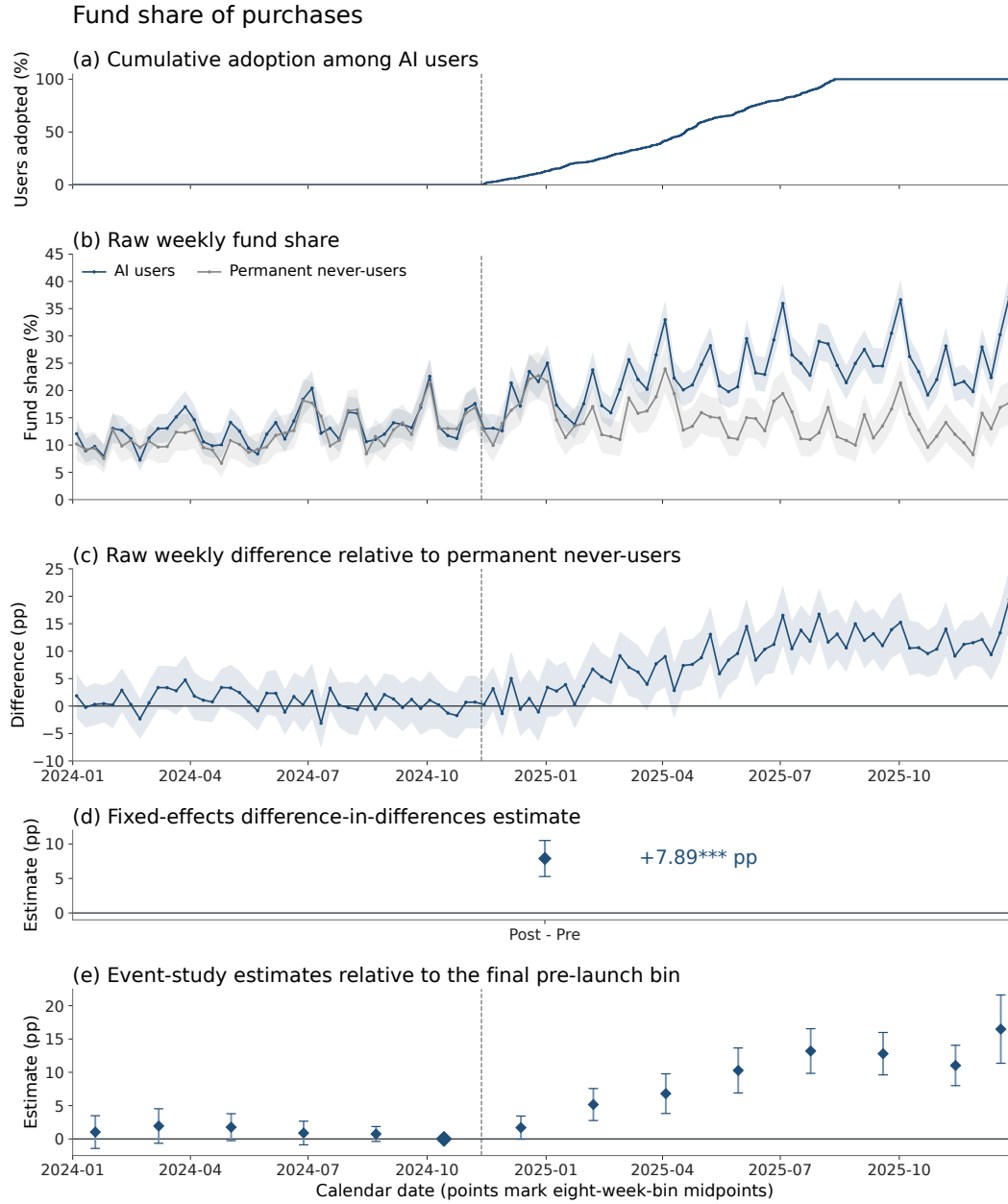


Figure D.5. AI introduction

Notes: The plot uses the November 2024 introduction of the AI assistant on Mezzi as the common start-of-treatment date (see Carlin et al., 2023, Figure 1). Panel A plots cumulative AI adoption. Panel B plots the weekly fund purchase shares for the two groups, and Panel C plots the difference between them. Panel D reports the DID estimates. Panel E reports event-study estimates in eight-week bins. All intervals are 95 percent. Panels D and E include individual and calendar-week fixed effects, with two-way clustering. The sample ends in December 2025.

Table D.7. Alternative Monthly Activity Requirements

Monthly purchase requirement	Point estimate	Standard error
Any positive purchase	6.80***	1.16
At least 0.1% of opening portfolio	5.78***	1.43
At least 0.5% of opening portfolio	7.60***	1.55

Notes: This table reports Sun–Abraham estimates using the activity restrictions indicated in each column. The dependent variable is the direct-index fund share of the monthly purchase portfolio. All specifications include individual and calendar-month fixed effects. Standard errors are double-clustered by individual and calendar month and reported in parentheses. Significance levels: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table D.8. Monthly Purchase Portfolio Effects across Adoption Cohorts

	Early	Middle	Late
<i>Panel A. Equal calendar-time periods</i>			
Post-adoption effect	4.83 (4.96)	7.70*** (2.36)	7.72*** (2.58)
Adoption dates	Oct. 2024– Mar. 2025	Apr.– Sept. 2025	Oct. 2025– Mar. 2026
<i>Panel B. Equal-number adopter groups</i>			
Post-adoption effect	6.09** (2.45)	7.78*** (3.01)	8.17*** (2.89)
Adoption dates	Nov. 13, 2024– July 17, 2025	July 17– Nov. 7, 2025	Nov. 8, 2025– Mar. 29, 2026

Notes: This table reports stacked difference-in-differences estimates separately by adoption cohort. The dependent variable is the direct-index fund share of the Monthly Purchase Portfolio. Standard errors are double-clustered by individual and calendar month and reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

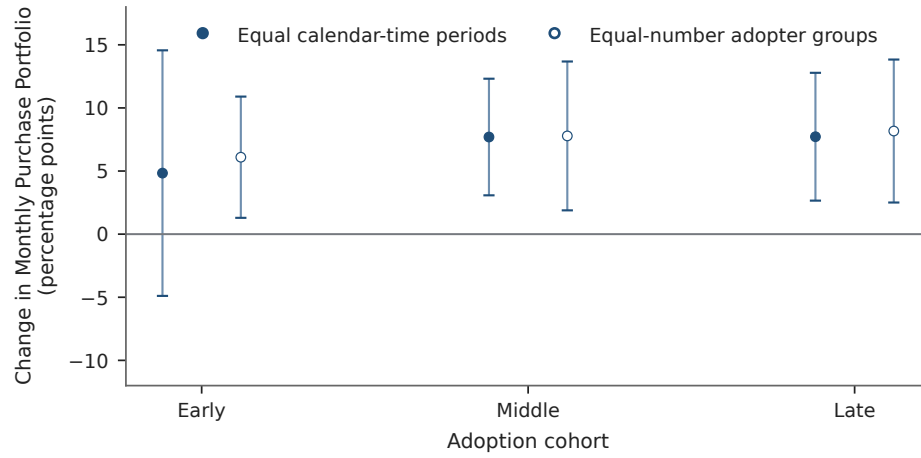


Figure D.6. Monthly Purchase Portfolio Effects across Adoption Cohorts

Notes: This figure plots the estimated coefficients and 95 percent confidence intervals from Table D.8. The dependent variable is the direct-index fund share of the Monthly Purchase Portfolio. Results are reported separately by adoption cohort.

E Recommendation-Classification Task

The following is an adapted excerpt from the prompt used to classify investment actions in initial AI answers. Omissions are indicated by [...].

```
Task: Classify the AI assistant's answer to a financial question. [...]
```

Identify if the answer contains financial advice, and if so what asset and if applicable what ticker the advice recommends, and whether it recommends to hold/buy/sell.

Return only one valid JSON object. Do not include markdown, code fences, explanations, comments, or any text before or after the JSON. JSON strings must be valid plain text and must not contain internal double quotes, backslashes, newlines, tabs, or control characters.

All categorical fields list a preferred set of values. If none fit, emit a short descriptive value of your choice in that field; it will be reviewed and normalized later. Use "unclear" only when the case is genuinely ambiguous.

```
[...]
atom_posture (preferred values):
- recommended_action: model recommends user take this.
- confirmed_action: model confirms a user-proposed or user-taken action.
- discouraged_action: model recommends not taking this.
- conditional_action: model recommends with timing or conditions.
- referenced_action: model mentions in analysis without recommending.
[...]
```

action_side (preferred values; use null for no action side):

```
- buy_add, sell_trim, [...]
```

```
[...]
object_code:
- If a ticker is named, use the uppercase ticker as object_code AND set symbol.
[...]
```

Rules:

```
[...]
3. When the answer presents alternative recommendations for the same goal or object, create
one atom per option with action_frame=option_set.
4. Items mentioned only as background context get atom_posture=context_mention with
appropriate context_role, action_side=null, action_frame=not_action.
[...]
```

15. If the answer contains no recommended or referenced action, emit empty answer_atoms with answer_posture=pure_information or declines_or_clarifies.

```
[...]
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F Additional Implementation Evidence

F.1 Recommended Portfolios Versus Alternative Purchases

Table F.1. Historical diversification and risk: recommendations versus purchases

	Main estimate	Winsorized 1/99	Investors weighted equally
Reduction in relative Sharpe-ratio loss	0.107*** (0.014)	0.106*** (0.013)	0.112*** (0.014)
Reduction in market beta	0.231*** (0.031)	0.235*** (0.031)	0.236*** (0.033)
Reduction in annualized volatility (pp)	11.27*** (1.37)	10.80*** (1.18)	12.22*** (1.38)
Reduction in maximum peak-to-trough decline (pp)	10.35*** (1.05)	10.33*** (1.03)	10.31*** (1.06)
Compared within the same advice event	Yes	Yes	Yes
Standard errors	Investor-clustered	Investor-clustered	Robust

Notes: Relative Sharpe-ratio loss follows Calvet et al. (2007): $1 - S_h/S_B$, where S_h is the relevant constructed portfolio Sharpe ratio and S_B is market Sharpe. Under the zero-alpha one-factor CAPM, it is the share of market Sharpe lost to imperfect diversification. The row reports the reduction in this loss, so positive values favor the recommendation. Both constructed portfolios use the same 60 months. The comparison is restricted to advice events for which both the constructed recommended and post-advice purchase portfolios have complete return histories. The main and winsorized columns weight comparisons equally and cluster standard errors by investor; the final column weights investors equally and reports robust standard errors. Maximum drawdown is the largest decline from a preceding peak. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table F.2. Characteristics of purchases that follow advice

	Follows advice vs. no recent advice	Follows advice vs. other post-advice purchase	Security named vs. other post-advice purchase
Diversified fund (pp)	31.56*** (3.03)	36.34*** (3.54)	24.88*** (4.57)
Individual stock (pp)	-33.10*** (3.23)	-40.86*** (3.66)	-29.44*** (5.76)
Broad passive equity (pp)	9.05*** (1.32)	10.66*** (1.49)	12.34*** (1.59)
Expense ratio (bp)	-6.17*** (0.82)	-9.09*** (1.15)	-7.64*** (1.27)
Historical volatility (pp)	-8.85*** (0.74)	-10.26*** (0.81)	-7.01*** (1.13)
Historical beta	-0.186*** (0.019)	-0.224*** (0.021)	-0.104*** (0.030)
Reduction in maximum peak-to-trough decline (pp)	8.50*** (0.70)	9.74*** (0.76)	6.68*** (1.08)
Individual FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes

Notes: Columns (1) and (2) compare purchases following advice with purchases without recent advice and other post-advice purchases, respectively. Column (3) compares named securities with other post-advice purchases. All specifications include investor and purchase-date fixed effects. The coefficients are measured in percentage points for composition and volatility, and basis points for fees. Standard errors are clustered by investor and calendar week and reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F.2 Broader Measure of Advised Portfolio Exposure

Final advised exposure rises much more strongly with interactions than with elapsed time

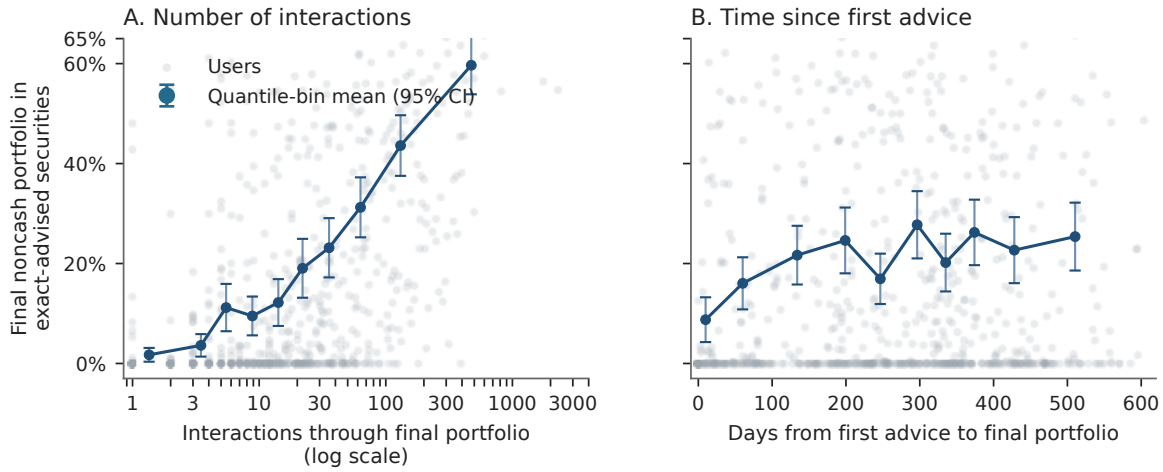


Figure F.1. AI-recommended securities in final portfolios

Notes: This figure shows the share of investors' final portfolios invested in securities previously recommended by AI. Panel A plots this share against the number of AI interactions on a log scale. Panel B reports the share invested against the days from the first AI recommendation. Both panels sort into deciles.

Table F.3. Advice implementation

	Trading days after the answer			
	1	2	5	10
<i>Panel A. Answer-level implementation</i>				
Constant (%)	16.6*** (2.15)	23.8*** (2.52)	32.0*** (2.59)	36.2*** (2.43)
R^2	0.000	0.000	0.000	0.000
<i>Panel B. Ticker-level implementation</i>				
Constant (%)	9.0*** (2.29)	12.6*** (2.60)	16.8*** (2.70)	18.5*** (2.46)
R^2	0.000	0.000	0.000	0.000
Individual fixed effects	No	No	No	No
Answer-date fixed effects	No	No	No	No
<i>Panel C. Ten-day regressions with fixed effects</i>				
	Pooled (1)	Individual FE (2)	Date FE (3)	Both FE (4)
Answer-level outcome mean (%)	36.2	36.2	36.2	36.2
Answer-level R^2	0.000	0.264	0.137	0.339
Ticker-level outcome mean (%)	18.5	18.5	18.5	18.5
Ticker-level R^2	0.000	0.223	0.151	0.281
Individual fixed effects	No	Yes	No	Yes
Answer-date fixed effects	No	No	Yes	Yes

Notes: Panels A and B report cumulative implementation rates (percent) from constant-only regressions. Implementation requires a purchase of any recommended security in an answer (A) or the specific ticker (B). Panel C adds the indicated fixed effects to the ten-day regressions. The sample consists of buy advice with ten trading days of follow-up. Standard errors, clustered by individual, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table F.4. Portfolio-detail personalization and subsequent behavior

	(1) Pooled	(2) Individual FE	(3) Date FE	(4) Individual + date FE
<i>Panel A. Exact-recommendation implementation within ten days</i>				
Account type or location	6.73*** (2.27)	3.86** (1.83)	4.61*** (1.67)	3.17** (1.49)
Holdings and allocation	4.82*** (1.70)	1.28 (0.96)	3.58*** (1.09)	1.16 (0.90)
Cash	4.14** (1.66)	3.92*** (1.14)	3.33*** (1.26)	4.01*** (1.02)
Concentration	0.93 (1.78)	1.01 (0.92)	-0.02 (1.27)	0.90 (1.09)
Transactions, gains, or losses	6.63** (2.67)	5.32*** (1.27)	3.89** (1.60)	3.54*** (1.13)
Outcome mean	36.25	36.25	36.25	36.25
R ²	0.014	0.254	0.137	0.326
	(1) Fund share of positive purchases	(2) Any eligible purchase week	(3) Closing fund holdings share	
<i>Panel B. Concentration detail and the post-advice portfolio response</i>				
Post × concentration detail	5.43** (2.50)	-2.15 (2.77)	-2.12** (1.00)	
Average post effect	2.64 (1.07)	8.63 (1.18)	-0.30 (0.42)	
Stack-individual fixed effects	Yes	Yes	Yes	
Stack-calendar-week fixed effects	Yes	Yes	Yes	
All detail-category interactions	Yes	Yes	Yes	

Notes: In Panel A, the dependent variable is an indicator for whether a recommended security is purchased during trading days 1–10 after the answer. All columns control for the eleven personalization categories; selected coefficients are shown. Panel B reports stacked difference-in-differences estimates. Estimates are in percentage points. Standard errors are clustered at the individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F.3 Recent Performance and Later Language

$$\text{Implemented}_{ia} = \alpha + \sum_{h \in \{5, 63, 252\}} \left[\beta_h R_{t-h:t-1}^{\text{VTI}} + \theta_h \left(R_{i,t-h:t-1}^P - R_{t-h:t-1}^{\text{VTI}} \right) \right] + X_i' \gamma + E_{ia}' \lambda + Q_{ia}' \delta + A_{ia}' \pi + \eta H_{ia} + \mu M_{ia} + \varepsilon_{ia}. \quad (4)$$

Here R^P is the investor's portfolio return and R^{VTI} is the market return. X_i , E_{ia} , Q_{ia} , and A_{ia} collect individual, prior-engagement, prompt, and recommendation-composition covariates; H_{ia} and M_{ia} are the shares already held and mentioned or implied by the prompt.

Table F.5. Recent performance and recommendation implementation

	Implementation within ten trading days	
	(1) Baseline covariates	(2) + Held/prompt shares
VTI return, prior 5 days (10 pp)	-17.43*** (6.03)	-15.41** (6.01)
Portfolio excess return vs. VTI, prior 5 days (10 pp)	-23.33*** (6.62)	-19.26*** (6.64)
VTI return, prior 63 days (10 pp)	4.33** (2.07)	4.82** (1.97)
Portfolio excess return vs. VTI, prior 63 days (10 pp)	2.75 (2.14)	3.30 (2.21)
VTI return, prior 252 days (10 pp)	4.01 (2.87)	4.61 (2.91)
Portfolio excess return vs. VTI, prior 252 days (10 pp)	3.37* (1.82)	3.05* (1.79)
Share of recommendations already held	—	36.65*** (2.66)
Share mentioned or implied by prompt	—	-0.29 (1.14)
Outcome mean (percent)	36.0	36.0
R^2	0.114	0.199
Held/prompt controls	No	Yes

Notes: The dependent variable is an indicator for whether a recommended security is purchased within ten trading days. All columns control for individual characteristics, prior AI engagement, and prompt and recommendation characteristics. Estimates are in percentage points; returns are measured in units of ten percentage points. Standard errors are clustered at the individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table F.6. Individual-stock portfolio performance and recommendation implementation

	Implementation within ten trading days	
	(1) Baseline covariates	(2) + Held/prompt shares
VTI return, prior 5 days (10 pp)	1.53 (5.37)	1.02 (5.04)
Individual-stock portfolio excess return, prior 5 days (10 pp)	-2.95 (3.64)	-2.51 (3.34)
VTI return, prior 63 days (10 pp)	1.75 (1.77)	1.67 (1.65)
Individual-stock portfolio excess return, prior 63 days (10 pp)	1.30 (0.91)	1.02 (0.91)
VTI return, prior 252 days (10 pp)	0.56 (1.92)	1.71 (1.83)
Individual-stock portfolio excess return, prior 252 days (10 pp)	0.32 (0.47)	0.22 (0.43)
Share of recommendations already held	–	37.07*** (2.84)
Share mentioned or implied by prompt	–	-0.43 (1.17)
Outcome mean (percent)	36.3	36.3
R^2	0.111	0.198
Held/prompt controls	No	Yes

Notes: This table repeats Table F.5 using the individual-stock portfolio return in place of R^P in equation (4). The sample is restricted to exact-buy recommendation responses from investors with a positive individual-stock portfolio on the day before the answer. I measure the individual-stock return as the value-weighted total return on positive individual-stock holdings. Column (2) adds the shares of recommendations already held and mentioned or implied by the prompt. The coefficients represent percentage-point changes in implementation for a 10-percentage-point higher return; share coefficients compare zero with one. Standard errors are clustered at the individual level and reported in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table F.7. Financial terms retained by the prevalence screen

Term	AI answers (%)	Individual prompts (%)	Ratio
Tax	47.4	9.4	5.1×
Cash	42.6	7.6	5.6×
Growth	42.0	3.5	12.0×
Risk	38.9	6.8	5.7×
Exposure	37.6	6.0	6.3×
Taxable	37.2	4.5	8.3×
Value	35.8	2.1	17.3×
Fund	33.9	4.2	8.0×
Allocation	31.3	6.1	5.1×
IRA	30.2	5.0	6.1×
Stock	28.0	4.6	6.1×
Equity	27.8	1.7	16.6×
Retirement	27.6	5.3	5.2×
ETF	27.5	2.5	10.8×
Sector	24.8	4.2	5.9×
Yield	21.4	1.6	13.1×
Volatility	20.6	0.6	36.4×
Dividend	20.1	4.2	4.8×
Index	20.1	0.9	23.1×
Bond	19.3	1.6	12.0×
Return	19.1	2.6	7.4×
Concentration	18.1	1.4	13.3×
Diversification	17.7	2.5	7.0×
Liquidity	13.5	0.6	24.5×
Roth	13.2	0.9	15.6×

Notes: This list only contains terms that appear in at least 10 percent of AI answers and at least three times as often in answers as in prompts. Frequencies refer to the word shown; terms are ordered by frequency in AI answers.

F.4 Advice-Compliant Versus Other Trades

Table F.8. Characteristics of advice-compliant and other signed trade baskets

	Advice-compliant trades	Other trades	Difference
Market beta	0.263 (0.075)	0.242 (0.053)	0.021 (0.080)
Annualized volatility (%)	19.88 (1.28)	13.99 (1.20)	5.88*** (1.16)
Idiosyncratic volatility (%)	10.34 (0.96)	7.87 (0.99)	2.47*** (0.85)
Signed-exposure HHI	0.197 (0.035)	0.117 (0.023)	0.079*** (0.029)
Relative Sharpe-ratio loss (%)	62.02 (5.30)	75.51 (7.11)	-13.49* (7.64)
Maximum drawdown (%)	34.13 (2.26)	24.94 (2.12)	9.18*** (2.16)
Net ETF/fund allocation (%)	48.89 (5.41)	24.55 (5.58)	24.34*** (6.45)
Net equity allocation (%)	31.23 (6.26)	21.50 (4.96)	9.74 (6.53)
ETF/fund share of purchases (%)	92.46 (2.04)	79.83 (4.62)	12.63*** (3.51)
Purchased-fund expense ratio (bp)	9.55 (1.62)	16.54 (4.52)	-6.99* (3.98)

Notes: This table compares actual advice-compliant trades with other trades by the same individual during trading days 1–10 after the same answer. Difference is defined as advice-compliant minus other. I weight advice events equally. Basket weights are purchases minus sales, divided by total purchases plus sales. Standard errors are clustered at the individual level and reported in parentheses.

I calculate return measures using a common 60-month return history. Both portfolios use fixed initial weights. Remaining cash earns the risk-free rate. HHI is $\sum_i w_i^2 H_i$, where H_i is the most recent fund-holdings HHI, or one for individual securities and uncovered funds; cross-fund overlaps are not combined. The purchased-fund expense ratio is the purchase-dollar-weighted average expense ratio of fee-covered funds bought. Purchase shares and expense ratios are calculated for buys only.

Relative Sharpe-ratio loss is $1 - \rho$, where ρ is the correlation between basket and market excess returns under the zero-alpha CAPM. For signed baskets, this measure can exceed one and is not a welfare comparison. These signed-trade comparisons do not measure changes in the risk of total portfolio holdings.

The sample requires common histories and complete paired outcomes for each measure; drawdown excludes nonpositive strategy wealth. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.